Lumentum (LITE)

Full investment research | Model assumptions and source evidence

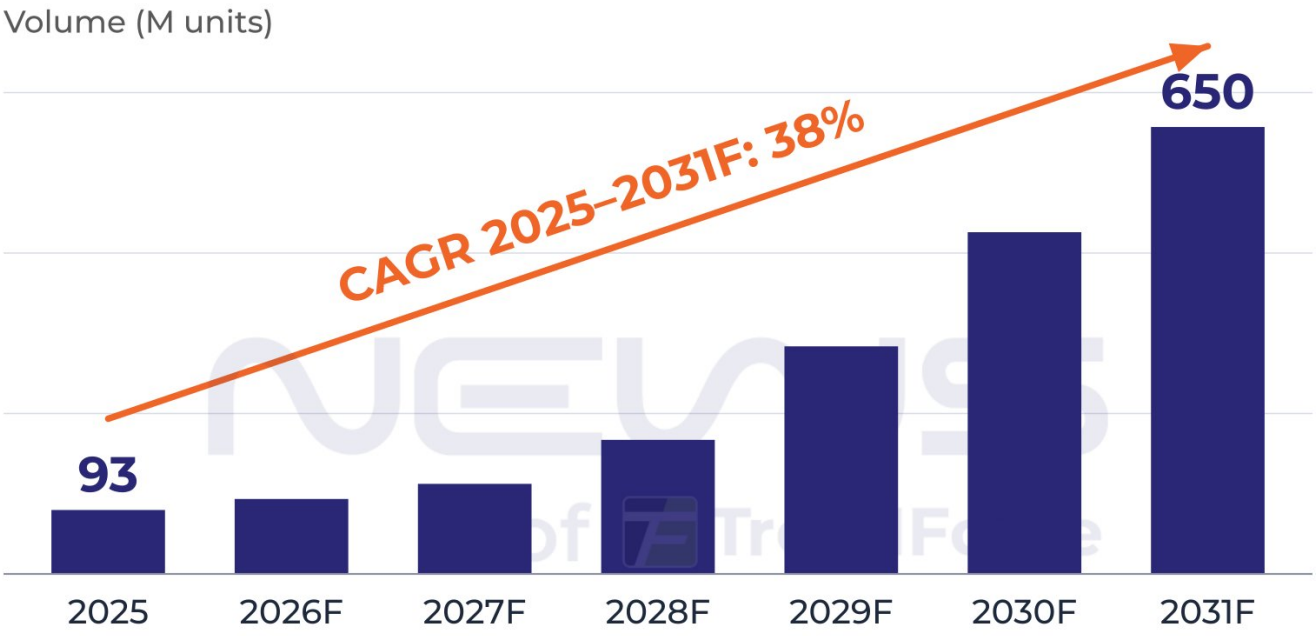
Early merchant VCSEL: a small separately scoped revenue case

Current Q4 2027 ceiling: $1,143.874580. The active forward-four-quarter operating profit is **$5,976.744682M**, valued at 20× and 104.5M diluted shares. The new VCSEL case contributes **$0.654516M operating profit** in that window, or **$0.125266/share**. It is a small LOW-confidence supplier-invoice case. Analytical Base 80 and protected Excel Base 99 remain unchanged.

TrendForce's October 8 release forecasts communications VCSEL output from 93M units in 2025 to 650M in 2031, including existing short-reach transceivers and NPO. It identifies early NPO product launches in 2027-28 and both 850 nm and 1060 nm routes. These are forecast device/output counts, **not complete optical engines**, orders or Lumentum revenue. The exact die-versus-array counting convention remains unresolved. The source's 38% CAGR is rounded. The intermediate bar shapes are not a constant-CAGR path.

Six Leading Suppliers Bet on VCSEL NPO; VCSEL Demand Projected to Reach 650 Million Units by 2031

Several NPO solutions expected to launch in 2027–2028



VCSEL-V1. Original TrendForce communications-VCSEL forecast, October8,2026, relayed in XDATA-ESA9. Exact printed endpoints are93M2025 and650M2031. The early2027/28 bars are approximate readings, not printed exact unit forecasts. Original exhibit, panel 1 of 2; read from top to bottom.



Source: TrendForce, "2027 Infrared Sensing and Optical Communication Market Trend," Oct. 2026

NEWS of TrendForce

VCSEL-V1. Original TrendForce communications-VCSEL forecast, October8,2026, relayed in XDATA-ESA9. Exact printed endpoints are93M2025 and650M2031. The early2027/28 bars are approximate readings, not printed exact unit forecasts. Original exhibit, panel 2 of 2; read from top to bottom.

Source/model quantity	2027	2028	Status and scope
Total communications VCSEL reported-unit equivalents,M	130	200	Model midpoint of source-image readings
Approximate visual reading bounds,M	120-140	180-210	Chart-read uncertainty; no invented precise annual forecast
Legacy short-reach equivalent baseline,M	102.532500	107.659125	93M 2025×1.05^(year-2025);5% growth is an assumption
Residual above legacy,M	27.467500	92.340875	Total less growing legacy; not all is automatically new NPO
Assumed NPO share of residual	50%	50%	LOW-confidence scope allocation
Incremental NPO equivalents,M	13.733750	46.170438	Residual×50%

A supplier route with qualifications and a distinct invoice

Lumentum's September 21 issuer demonstration describes a 1060 nm VCSEL/photodetector optical engine with Qualcomm and Corning, using high-density two-dimensional arrays and backside optics. Its 32 Gb/s NRZ demonstration and combined transmit/receive bandwidth support a technical route, not a commercial unit award or price. The scale-up application and existing high-volume 3D-sensing manufacturing make a small early ramp plausible. Reach, coupling/dispersion/noise, temperature, customer acceptance and the higher 1060 nm cost remain realization constraints.

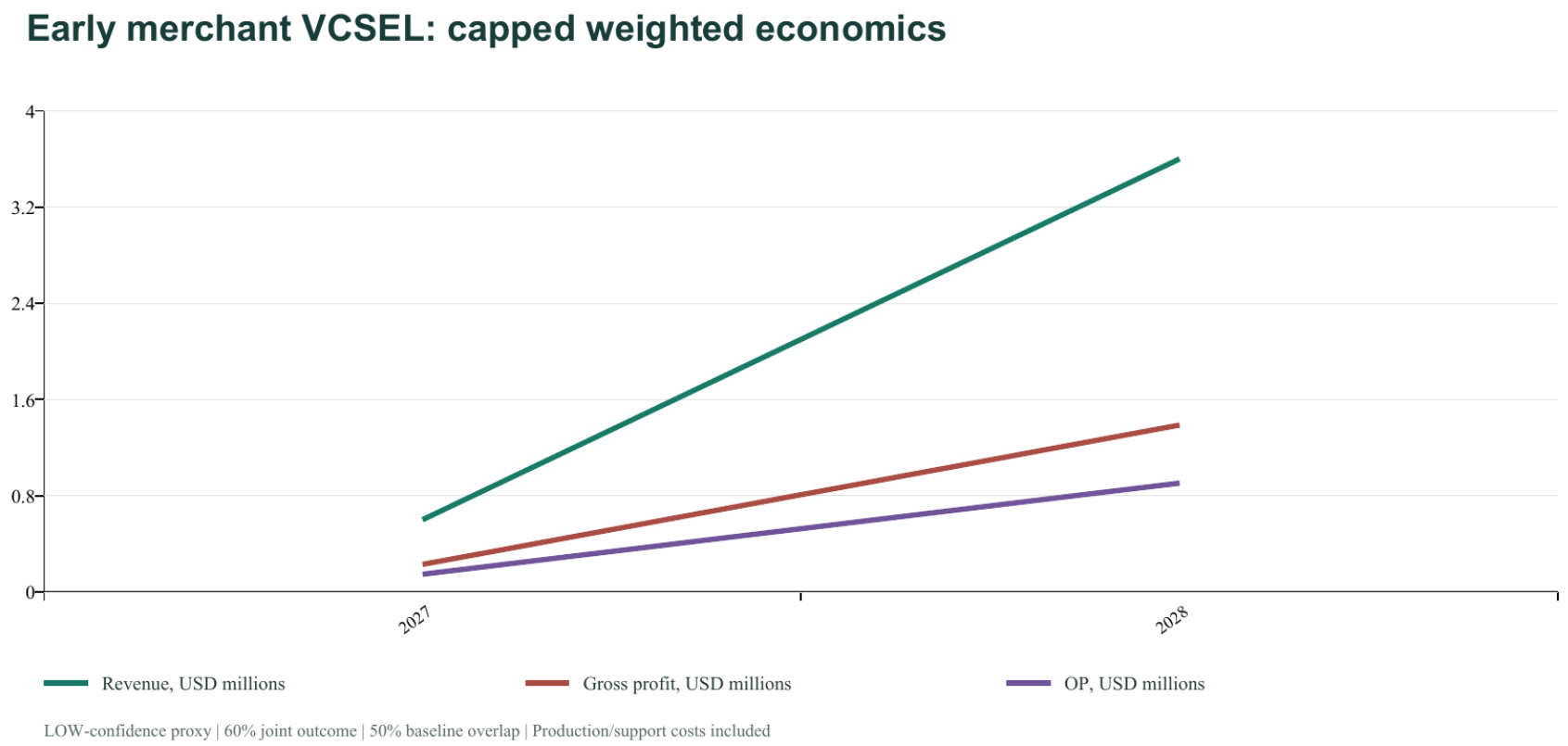
ASTRA selects 60% for the joint small early NPO-adoption, Lumentum qualification and externally invoiced outcome. This is subjective LOW-confidence model judgment, retaining the prior adoption weight while explicitly making supplier realization part of the state. A global adoption probability does not independently prove Lumentum sales. The 30% OCS,40% FCC and 55% UHP/ELS economic probabilities remain separate and unchanged. This GaAs VCSEL invoice is not another InP UHP/ELS premium.

Driver	Selected assumption	Boundary
Lumentum capture	10% of allocated NPO equivalent units	Assumption across suppliers, before external-invoice fraction
Externally invoiced fraction	75%	Removes assumed captive/internal content; no second engine invoice
Qualified accepted output cap	0.5M 2027;3M 2028	LOW-confidence reported-unit-equivalent cap, not verified physical device/array capacity
Matching invoice ASP	$4/equivalent; diagnostic$1-8	One matching unit invoice is an explicit inference; not a reported price or InP/engine ASP
Baseline economic overlap	50%	Partial allocation against existing Other growth and displacement; not proof that exact new cohort is already modeled

Driver	Selected assumption	Boundary
Cost pool	Current Other gross margin and quarter-specific group reference opex	Avoids importing high InP scarcity margins
Extra costs	3 pp manufacturing/qualification/D&A plus 3 pp qualification/support opex	Assumptions on incremental sales; existing fixed costs and depreciation remain
Recognition phases	2027:0/0/25/75%;2028:10/20/30/40%	Calendar Q1-Q4; existing fiscal labels retained

The two 50% deductions have different denominators: NPO mix applies to the market residual; baseline overlap applies to the gross merchant revenue. Qualified accepted caps already describe output after qualification, so there is no second good-unit-yield multiplication. The baseline Other line covers 3D sensing, telecom and other components and rises from$265M calendar 2027 to$310M 2028. That broad growth motivates partial overlap but does not identify 100% of this particular device cohort. The 50% joint net reserve also covers potential displacement/cannibalization of legacy products and prior growth; it does not prove costless spare capacity or a fully incremental customer wallet. No second 50% haircut is applied. The small accepted-output caps assume existing VCSEL capability and the current capacity plan can serve the early case, with the stated production/D&A and support buffers. They establish no spare-capacity fact, second factory, cash-capex benefit or new capacity funding. The current 2026/27 product growth and all UHP/ELS,OCS,FCC allocations remain intact.

Unit-to-invoice and cost bridge



VCSEL-V2. Selected early merchant VCSEL financial contribution, USD millions. One60%-weighted,50%-overlap case; cost buffers included. Model assumptions, not guidance.

Calendar year	Accepted equivalents,M	Gross merchant revenue	Conditional revenue after overlap	Weighted revenue	Weighted GP	Weighted opex	Weighted OP
2027	0.500000	2.000000	1.000000	0.600000	0.228600	0.081750	0.146850
2028	3.000000	12.000000	6.000000	3.600000	1.387080	0.482688	0.904392

For each year, external potential units = incremental NPO residual×10% capture×75% external invoice. Accepted output is the lesser of that result and the explicit cap. Both caps bind in the selected midpoint case. Gross sales = accepted equivalents×$4. Active sales = gross sales×(1-50% overlap)×60%. Each quarter uses its phase and current Other GM; incremental GP = active sales×(Other GM-3 pp), and incremental opex = active sales×(current reference opex+3 pp). The two cost buffers reduce operating profit by 6 pp of incremental sales.

Calendar quarter	Phase	Weighted revenue	Incremental COGS	Incremental GP	Incremental opex	Incremental OP
Q1 2027	0%	0.000000	0.000000	0.000000	0.000000	0.000000
Q2 2027	0%	0.000000	0.000000	0.000000	0.000000	0.000000
Q3 2027	25%	0.150000	0.093300	0.056700	0.020550	0.036150
Q4 2027	75%	0.450000	0.278100	0.171900	0.061200	0.110700
Q1 2028	10%	0.360000	0.221400	0.138600	0.048672	0.089928
Q2 2028	20%	0.720000	0.441360	0.278640	0.096912	0.181728
Q3 2028	30%	1.080000	0.663120	0.416880	0.144720	0.272160
Q4 2028	40%	1.440000	0.887040	0.552960	0.192384	0.360576

Calendar 2027 Q1-Q4 maps to Lumentum fiscal FQ3FY2027,FQ4FY2027,FQ1FY2028,FQ2FY2028;2028 maps similarly to the next fiscal year. No increment is modeled in 2026. No additional 2029+ branch is extrapolated without an economic timing/continuation bridge; this is a modeling scope choice, not an assertion that TrendForce's forecast stops in 2028.

Probability and unit-price sensitivity

Only the 60%/$4 case is active. The diagnostic changes below hold accepted caps, market scope, capture, external fraction, overlap, phases, quarter-specific margins, cost buffers, shares and 20× method constant.

Changed probability	Q4 2027 ceiling USD/share
0%	1,143.749314
30%	1,143.811947
60%	1,143.874580
100%	1,143.958091

Changed matching unit-equivalent ASP	Q4 2027 ceiling USD/share
$1	1,143.780631
$4	1,143.874580
$8	1,143.999847

The no-increment control removes only this new VCSEL branch and reproduces every existing quarterly ceiling. Full baseline overlap also removes it. The price diagnostic is conditional on the explicit device-equivalent counting convention: a different array/die definition must change unit count and matching ASP coherently,

rather than inflating both.

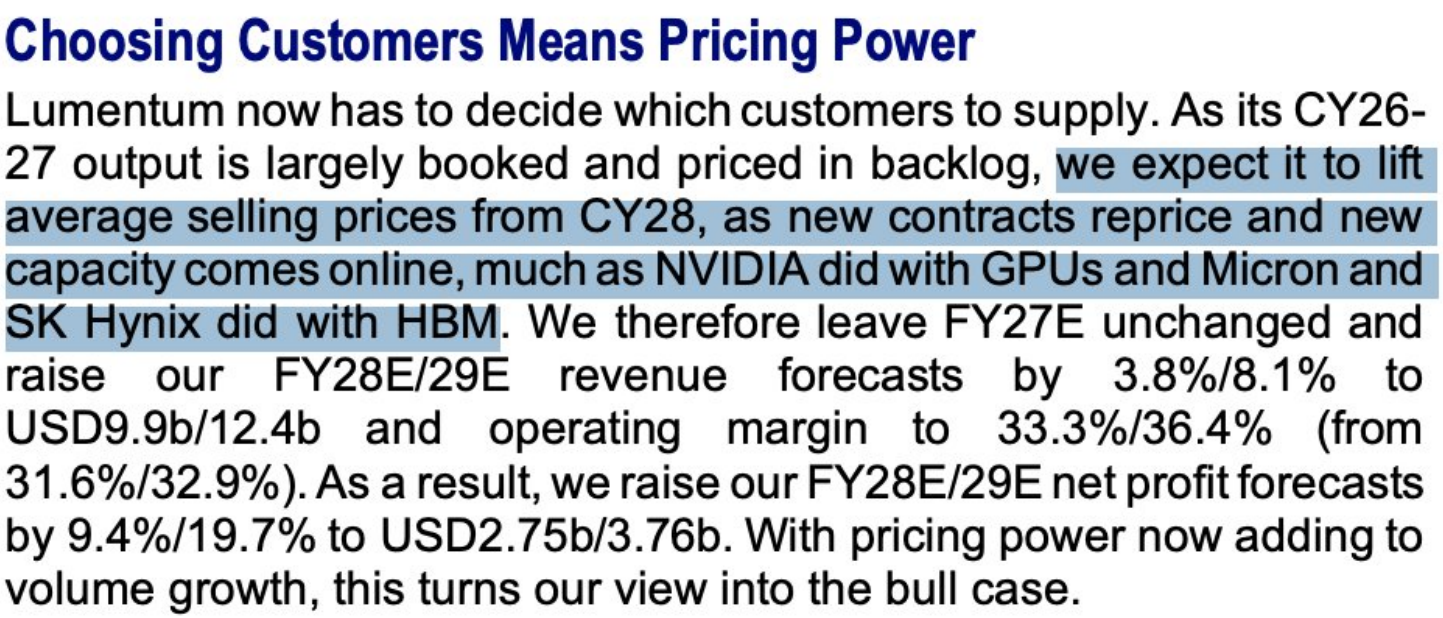
Revision triggers: raise or replace proxies when relevant supplier/customer evidence identifies accepted merchant units, matched device-versus-array invoice, customer allocation, realized ASP, production costs and timing. Reduce/remove the branch on qualification failure, delivery slips, captive-only billing, costs above the buffer, or a demonstrated identical cohort already inside the reference. Specific legacy-replacement margins, displaced units and qualified available-capacity evidence can change the joint net reserve and accepted cap. Lack of official revenue alone is not a reason to remove a forward case. This update adds no new candidate, second factory/capex bonus, royalty stream or product-margin uplift to the remainder of the model.

Source ledger: signal VCSEL-FORECAST; source releaseOctober 8 and issuer demonstrationSeptember 21; processedOctober 12. XDATA-20261011-1252-ET-ESA9/R1 posts 2108226117542768685 and 2108232083793723745; chart image/post 2108195712437735681. Mandarin source and English source describe the same forecast, not independent confirmations. Original imageSHA-256ec684c303bec87b3b3d8fd55c2027e7388d18862b7526750ef4e6856b563ddbc. Full file hashes are retained in the synchronized model JSON. Counting/ASP and qualified-output caps remain explicit assumptions.

Contract timing and the active 2027 premium

The current Q4 2027 ceiling is **$1,143.874580**, on the sanctioned20× forward-four-quarter operating-profit basis and104.5M diluted shares. Analytical Base is **80** and protected Portfolio Excel Base is **99**.

An October 11 post attributes an excerpt to Globlex Securities: much of calendar 2026-2027 output is already booked and priced, with broader ASP improvement from calendar 2028 as new contracts and capacity arrive. The screenshot does not show a broker header, original report date, SKU mix or customer contract terms. Its fiscal FY2028/FY2029 revenue and profit forecasts are not imported into the calendar operating model. The later reply explicitly says the size of any ASP increase is unknown. LITE-CT1 LITE-CT2



Original broker excerpt relayed October 11. The calendar contract-timing statement is attributed evidence; original report identity/date and product scope remain unverified.

ASTRA timing judgment: 30% probability that the existing 2027 eligible premium cohort cannot reprice until 2028. The alternative 70% retains the already narrow 2027 renewal/new-order pool because the aggregate broker claim does not identify these UHP/ELS contracts. Keep the original **55% economic probability**, **5% conditional premium**, **50% eligible share**, **50% economic-overlap reserve**, and **10% incremental COGS plus 5% incremental operating expense**. This weights the prior 2027 phases by 70%; 2028 onward is unchanged and receives no deferred-revenue catch-up. The effective joint weight for the affected 2027 pricing outcome is 55% ×70% = 38.5%, before eligibility, overlap and phase. This is an assumption about contract timing, not proof of a fixed-price clause or a new earnings release.

Quarter	Active premium phase
Q1 2027	17.5%
Q2 2027	35.0%
Q3 2027	52.5%
Q4 2027	70.0%

The Q4 2027-Q3 2028 reference UHP/ELS pool remains **$1,225M**. Only its $235M Q4 2027 portion receives the 70% timing weight, giving a timing-adjusted pool of **$1,154.5M**. The active price increment is **$7.937188M revenue**, less **$0.793719M COGS** and **$0.396859M operating expense**, or **$6.746609M OP**. That retains

$1.2912/share of scarcity value. 411984M OP **and** $0.078849/share**.

The original scarcity evidence and this timing counter-evidence concern the same pricing pool. Neither creates a new layer of unit growth, nor reprices FCC-generated sales, CW/EML or OCS. Contract/SKU coverage, accepted reset terms, first shipment dates and incremental support costs are the next discriminating evidence. Fixed-price commitments and supply scarcity can coexist.

Current source review, including the October 11 contract-timing assessment and October 12 VCSEL economics: September conference claims remain integrated with source images and explicit overlap/cost treatment. The current tables below include the later timing reserve; Cisco alone adds no separate economics. Analytical Base remains **80/100**; the owner-reinstated Portfolio Excel Base is **99/100**.

October 8: Cisco's laser-supply constraint supports the existing pricing case

The supplied **Cisco1.jpg and Cisco2.jpg** are consecutive conference takeaways attributed to meetings with Cisco investor-relations director Emily Hunt. Cisco's own schedule confirms her September 23, 2026 appearance at J.P. Morgan's U.S. All Stars Conference in London. A matching public index identifies a September 28 J.P. Morgan note by Joseph Cardoso, but the source crops omit the original header; the note date and author remain provisional. This is September evidence reviewed on October 8, not a new October price agreement. [LITE-CF1, CF2]

A customer identifies laser scarcity, but does not identify Lumentum's invoice

The note describes Cisco's supply environment as tight, with optics and particularly lasers the most acute constraints. Cisco says it has raised inventory, used advance commitments and LTAs, and diversified supply where practical. It also says its FY2027 guide is supported by available supply, with some capacity for modest upside. Both statements matter: supplier scarcity can support pricing, while Cisco's preparedness limits any inference of an immediate shipment crisis.

substantially more bandwidth/content per GPU, as each GPU must communicate coherently with extremely low packet loss.

- **Aging switches and security needs drive enterprise opportunity.** The enterprise opportunity is being driven by both aging installed equipment and an increasing security imperative. On that note, roughly $40 bn of campus switches are approaching last day of support, while the broader networking installed base approaching end of life represents a $100 bn opportunity over several years. For example, Catalyst 4000 is expected to reach end of support at the end of CY26, with Catalyst 6000 following the next year. While Cisco acknowledged increasing competition for its installed base, it noted that the competition will likely be at the device level, as Cisco targets large enterprises undertaking broader architectural changes that encompass networking, security, observability, and software.
- **Security and observability growth lagged ambitions but showing signs of turning.** Security/Observability was acknowledged as falling well short of Cisco's growth ambitions, with management attributing this partly to market complexity and partly to portfolio transition, such as Splunk customers moving toward cloud subscription. That said, the newer, refreshed portfolio was highlighted as growing better, and Cisco believes security has now turned a corner, with an expectation of exiting FY27 at high-single digit growth.
- **Supply framed as tight but not constraining FY27.** The supply environment for Cisco was described as tight rather than outright constrained, as the company has raised inventory, placed advance-purchase commitments, entered into long-term agreements, and is actively working through long-term silicon requirements. The most acute tightness appears to be in optics, particularly lasers. On the flip side, Cisco has reduced memory requirements in some products and is using supplier diversification where practical. Importantly, the FY27 guide is not supply-constrained, as it believes it has the supply needed to support demand, with some buffer for upside should demand track modestly higher.
- **Hardware growth weighs on gross margins "modestly" and drivers extend beyond hyperscaler mix alone.**The gross margin decline was characterized as "modest" in FY27, driven by hardware growing faster than higher-margin portions of the portfolio such as software, security, and services. That said, Cisco underscored that the mix issue is broader than hyperscaler mix, despite investor perception, as the company is selling more hardware across enterprise as well, with software and service trends lagging the hardware deployment and recognized over time.

Original supplied Cisco2.jpg: Cisco supply and gross-margin commentary. The company identifies optics/laser tightness while saying FY2027 guidance is adequately supplied. This does not identify a Lumentum allocation or price award.

Cisco's issuer release independently projects **$7.5B of hyperscaler AI infrastructure revenue in FY2027**, compared with approximately $4B in FY2026. The conference note's 60% Silicon One systems/40% optics split describes hyperscaler **orders**. Neither measure gives Lumentum a disclosed supplier allocation or turns complete Cisco optics revenue into laser revenue. Cisco's willingness to reserve supply is useful customer-side evidence; the exact laser SKU, supplier, qualified quantity, accepted ASP and contract terms still determine Lumentum's profit. [LITE-CF3]

The scarcity economics already receive a nonzero active weight

Retain **55% economic probability** for qualified UHP/ELS repricing, with the October 11 timing reserve applied only to the defined 2027 cohort; 2028-29 phases are unchanged. The bounded case remains a **5% conditional premium on 50% of eligible new or repriceable sales**, after a **50% overlap deduction** against the original reference economics. Cisco's new source adds **100% evidence overlap** with that existing branch; that second overlap does not remove the branch's active profit. It prevents the same shortage being priced twice.

Q4 2027-Q3 2028 scarcity bridge	USD millions or specified rate
Existing pre-premium UHP/ELS revenue pool	1,225.000000
Conditional price premium	5%
New/repriceable share	50%
Incremental share after existing economic overlap	50%
Timing-adjusted UHP/ELS revenue pool	1,154.500000
Conditional incremental revenue after timing reserve	14.431250
Selected outcome probability	55%
Active probability-weighted incremental revenue	7.937188
Additional manufacturing/support cost, 10% of incremental revenue	0.793719
Additional operating expense, 5% of incremental revenue	0.396859
Active probability-weighted operating profit already included	6.746609
Additional revenue, COGS, opex and OP from the Cisco crop	0.000000 each

The current calculation is ($235M x 70% + $990M) x 5% x 50% x 50% x 55% = $7.937188M of active incremental revenue. After the stated 15% combined extra cost, $6.746609M of OP is already included, worth approximately $1.2912 per share at 20x and 104.5M shares. The source supplies no additional Lumentum good-unit capacity, no contract repricing beyond this defined case and no independent new product pool. No fresh OCS customer award or change to the existing 40% FCC and 30% OCS branches follows.

Contracts can protect demand while limiting repricing

Advance purchase commitments and long-term supply agreements can reserve scarce output and improve utilization visibility. They may also fix prices or impose supply obligations. Evaluate premiums on new or repriceable quantities separately from fixed-price backlog, and include qualification, support, working-capital and capacity costs. A customer order window is also not a shipment schedule: the note says Cisco counts only orders expected within 90 days in reported bookings even though hyperscalers can order six to nine months before delivery.

The concurrent Coherent conference crops describe customer-funded capacity and take-or-pay structures. That is a competitor execution and capital-efficiency consideration, not evidence that Lumentum's supply advantage disappears. NVIDIA's cited $2B Coherent funding was a March common-equity investment, separately accompanied by a product purchase commitment; it is not generic prepayment accounting or a fresh October transaction. It does not change Lumentum's own shares, treasury or forecast. [LITE-CF4]

The useful next evidence is product-specific: executable prices, new versus fixed commitments, qualified high-power laser output, supplier awards, accepted delivery dates and associated manufacturing costs. General laser scarcity alone does not resolve the particular UHP noise/power qualification debate, and every Cisco optical deployment need not use Lumentum UHP devices.

Q4 2027 remains $1,143.874580 per share at the sanctioned 20x forward-four-quarter operating-profit method and 104.5M economic diluted shares. Analytical Base remains 80 and protected Portfolio Excel Base remains 99. Existing financial paths and nonzero scenario contributions remain intact; this additional source produces no new numerical increment.

Sources: **LITE-CF1:** original supplied Cisco1/Cisco2 conference crops. **CF2:** Cisco September conference schedule and matching secondary note metadata, provisional. **CF3:** Cisco August 12 FY2026 earnings and FY2027 outlook. **CF4:** Coherent March 2 equity financing 8-K.

Marvell Investor Day and engineering integration: October 6, 2026. The current research integrates the optical-engine roadmap, external-source architecture, fixed-wavelength product mapping and named Marvell-Lumentum OCS interoperability. Existing financial paths and selected probabilities reconcile to the active model; supplier capture and qualification remain product-specific.

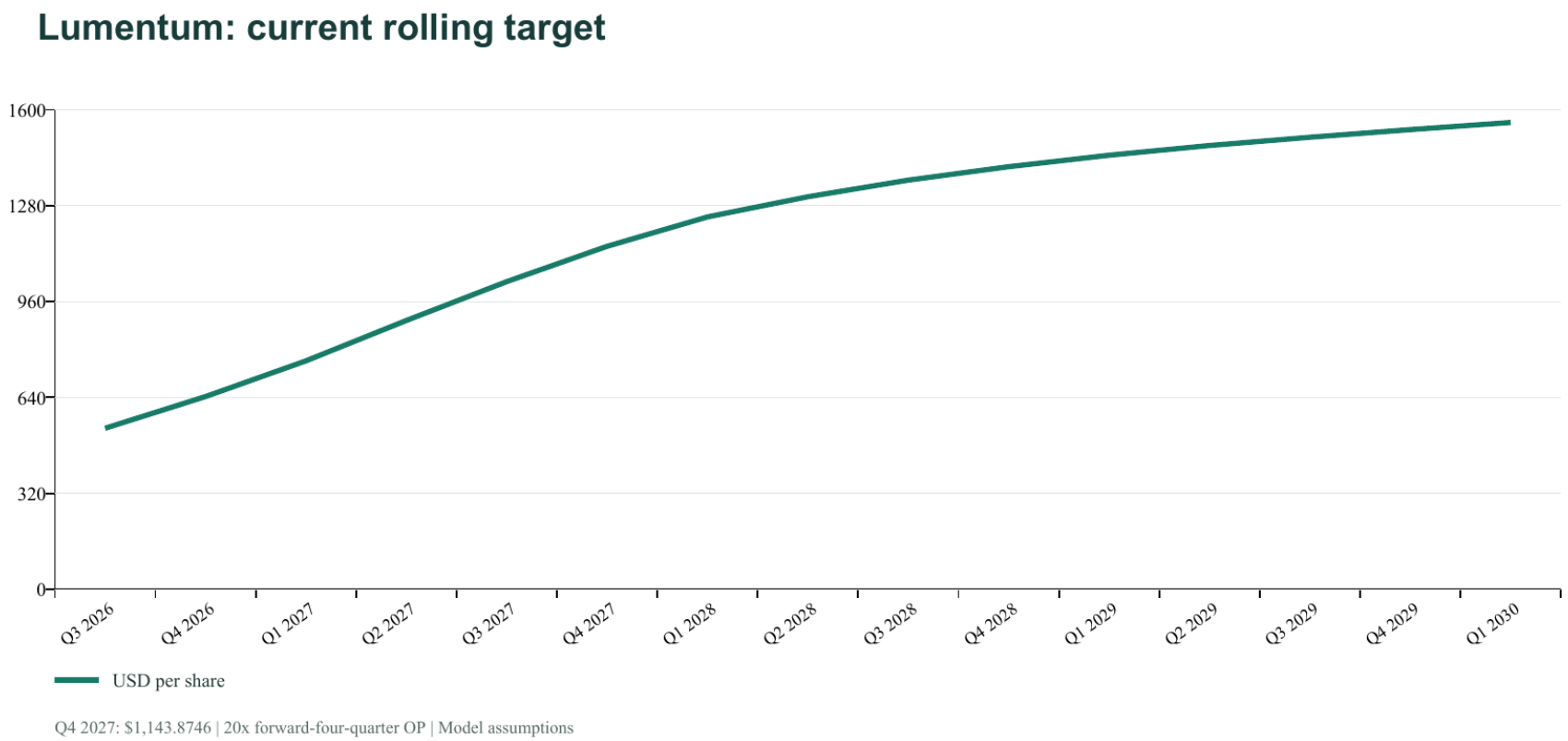
CURRENT EDITION - 12 October 2026 | Framework 8.0

Full investment research and current probability-weighted operating model. Financial assumptions, source evidence and forward estimates are distinguished throughout.

Q4 2027 price target

Current model item	Value	
Q4 2027 price target		1,143.8746 USD/share
Valuation window	Q4 2027 through Q3 2028	
Artifact analytical Base		80/100
Portfolio Excel Base		99/100, owner override
Method	20x forward-four-quarter operating profit; existing shares	

The active case includes a bounded, probability-weighted scarcity premium on UHP laser and ELS sales, plus a separate capped early merchant VCSEL contribution in Other components. The model also includes its 30% OCS scenario and 40% FCC scenario, with explicit product volumes, costs and diluted-share schedules. The next horizon roll is January 1, 2027.



Lumentum active price target path, USD/share. Q4 2027 is noted below; one current model, not company guidance.

price target quarter	Current USD/share
Q3 2026	536.8790
Q4 2026	642.5752
Q1 2027	761.6683
Q2 2027	896.3434

price target quarter	Current USD/share
Q3 2027	1,026.0066
Q4 2027	1,143.8746
Q1 2028	1,241.9121
Q2 2028	1,309.2214
Q3 2028	1,364.5844
Q4 2028	1,409.4216
Q1 2029	1,447.6964
Q2 2029	1,480.0628
Q3 2029	1,507.7297
Q4 2029	1,533.1117
Q1 2030	1,556.9111

October 7 review: laser supply talks and the dated share register

P Equity Research's October 6 post says Lumentum is negotiating long-term laser-supply agreements covering **calendar 2028 and 2029**, with potential additional pricing. This is an attributed, unconfirmed commercial breadcrumb. The attached image is a Lumentum logo; it supplies no contract terms, independent bank attribution or qualification evidence. Negotiations are distinct from executed quantities, accepted prices and delivered product. [LITE-LTA1]

The economic mechanism is already in the active model

Retain the **55% economic probability of qualified repricing**, with the current 30% timing-delay risk applied only in 2027, and the existing bounded **5% conditional premium on 50% of eligible UHP/ELS sales**, after the prior **50% overlap deduction** against reference economics. In the active Q4 2027-Q3 2028 window, the pre-premium UHP/ELS pool is $1,225M. After the October 11 timing adjustment, the calculation is ($235M ×70% + $990M) ×5% ×50% ×50% = **$14.43125M conditional revenue**, or **$7.937188M** after 55% weighting. Incremental manufacturing/support cost of 10% and operating expense of 5% leave **$6.746609M weighted OP**, already worth about **$1.2912 per share** at 20× and 104.5M shares.

The latest negotiations concern the same scarcity, repricing window and product pool. Their **additional evidence overlap is 100% with that existing branch**. This second overlap statement does not remove the existing contribution; it prevents adding it again. That October 7 source does not separately change the outcome probability, premium, capacity or costs; the current timing phases include the later October 11 reserve. The post does not provide an independent OCS customer wallet or an additional FCC qualification benefit. Both separate existing branches remain unchanged. Additional revenue, COGS, opex and OP from this particular post are therefore **zero**, because its mechanism is already modeled, not because unconfirmed evidence is excluded.

The future evidence that could change the model is specific: executable price terms for new/repriceable units, product scope, qualified good-unit capacity, delivery/acceptance dates and associated costs. A long-term agreement can protect demand while also fixing a price or reserving scarce capacity. Its duration alone does not prove another premium.

Shares outstanding are not the forward economic denominator

The October 6 proxy reports **90.582856M common shares at September 24**, plus **2.876415M preferred shares convertible one-for-one**, equivalent to **93.459271M** common shares before other dilutive instruments. The preferred voting rights differ for director elections. The August 28 common-share observation was **90.385389M**. The 0.197467M change between those common-share dates is a dated reconciliation, not an unmodeled capital raise by itself. [LITE-PROXY1,ARS1]

The model's **104.5M Q4 2027 economic diluted shares** include a different forward instrument/time basis. Replacing them with either the latest basic common count or the simple common-plus-preferred total would omit potential dilution. Conversely, adding the entire common-share increase, all outstanding awards or the unused incentive-plan reserve on top of 104.5M could count instruments twice. The proxy's 2.943955M outstanding awards and 3.478694M future plan reserve are not equivalent to immediate new shares. The reserve is authorization, not an issued economic claim.

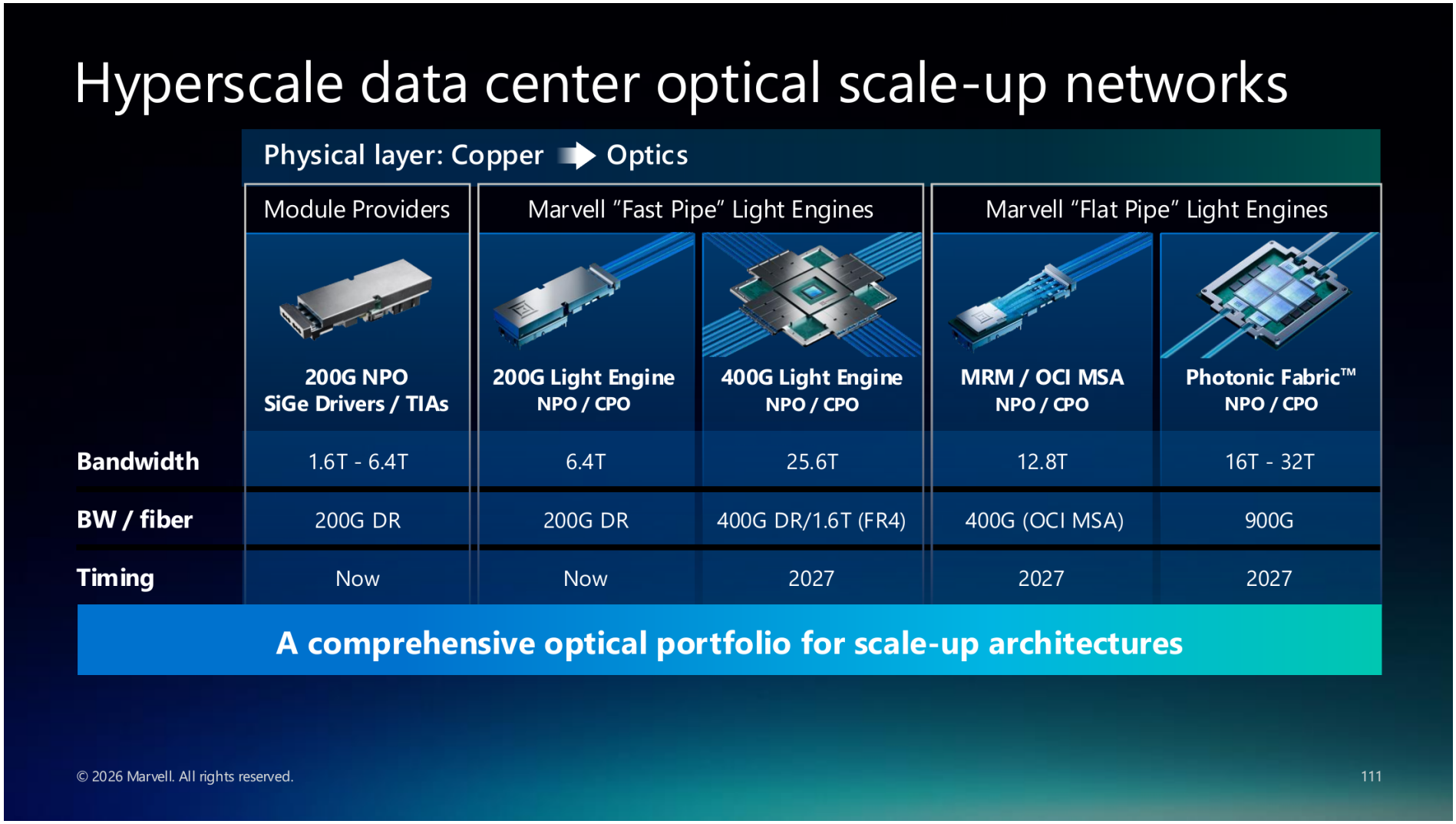
The old 77.80M basic-share statement in the historical calibration section below is now explicitly dated as superseded. The proxy improves the reconciliation but does not establish incremental dilution beyond the retained forward schedule. No new share issuance, treasury credit or plan-reserve dilution is added. **Q4 2027 remains $1,143.874580 at 20× F4Q OP**, with analytical Base **80** and Portfolio Excel Base **99**. Conversion, settlement and award details remain the inputs required to revise that forward economic share bridge coherently.

Marvell roadmap and Lumentum revenue capture

Marvell's October 6 Investor Day strengthens the engineering explanation for Lumentum's opportunity. Its roadmap moves optical links closer to accelerators and into larger scale-up domains. This expands the places where an efficient, stable and manufacturable light source matters. The conclusion is product-specific: more optical bandwidth can support more Lumentum revenue, but a bandwidth label does not specify the number of laser dies, their price or the supplier's share.

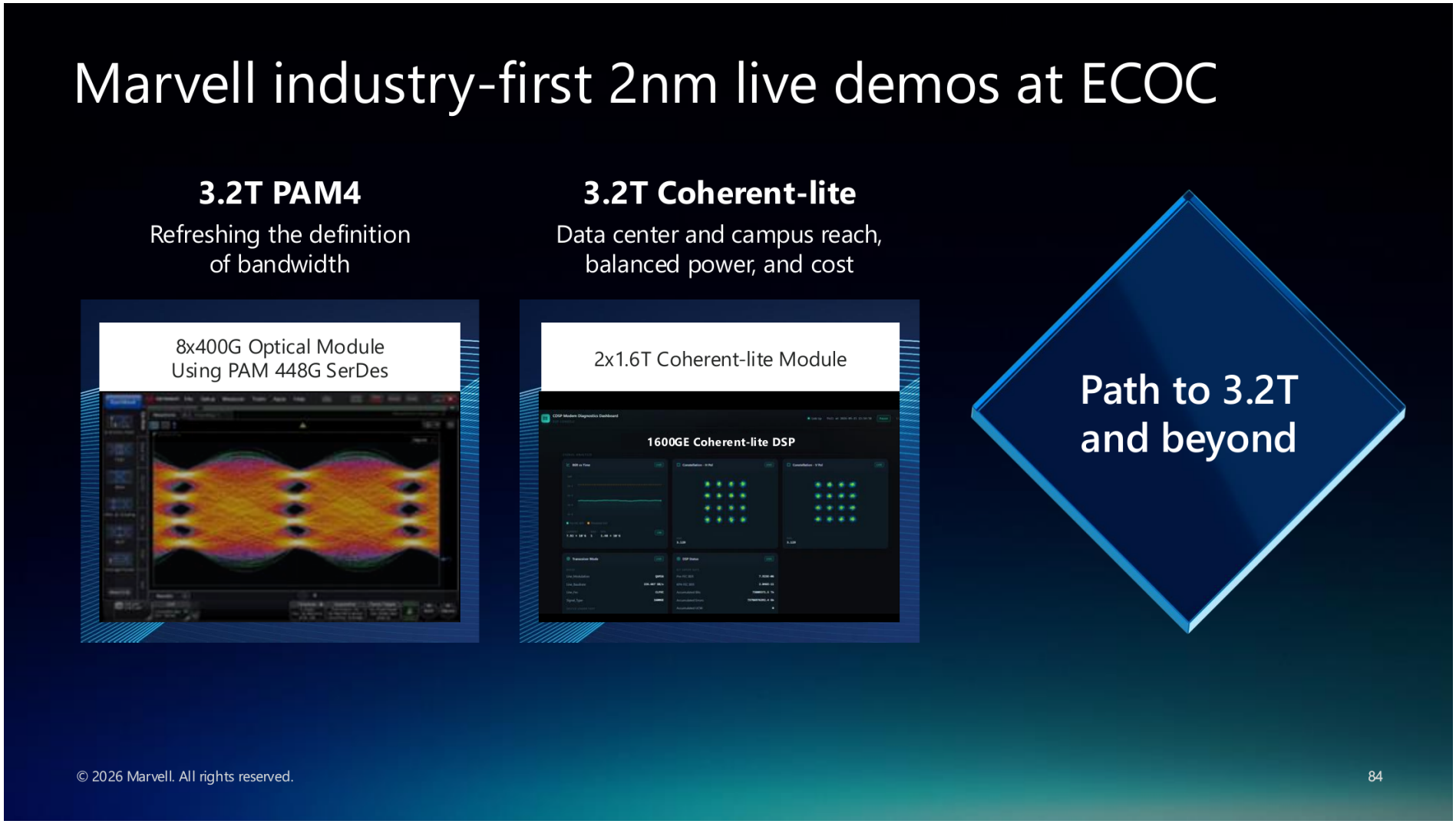
What the roadmap establishes

Page 111 separates several paths. Merchant module providers and Marvell's 200G Fast Pipe engine are labeled available now; Marvell places its 400G Fast Pipe, OCI/MRM and Photonic Fabric engines in 2027. These products differ in aggregate bandwidth, bandwidth per fiber and signal processing. The dates describe a product roadmap, not accepted customer shipments. They are consistent with Lumentum's existing model of ELS deliveries in the second half of 2027 ahead of scale-up deployments in 2028. They do not establish an earlier Lumentum ramp.



Marvell Investor Day, October 6, 2026, PDF page 111. Aggregate engine bandwidth, bandwidth per fiber and roadmap timing are distinct. Now/2027 is a Marvell product label, not Lumentum accepted sales or an earlier LITE ELS delivery date.

Page 84 also distinguishes two demonstrations that should not be mixed. The PAM4 module uses eight 400G optical lanes. The coherent-lite module combines two 1.6T paths to make 3.2T in aggregate. It is not evidence of one 3.2T coherent lane. Marvell's September 20 ECOC announcement describes the underlying 2nm demonstrations; the October deck is a clearer presentation of that event, not another independent demand forecast. [LM1]



Marvell Investor Day, PDF page 84. The ECOC demonstrations show 3.2T aggregate bandwidth as 8x400G PAM or 2x1.6T coherent-lite. Neither is a Lumentum laser bill of materials, and a live link demonstration does not establish production yield or volume.

Why source quality matters more than a headline wattage

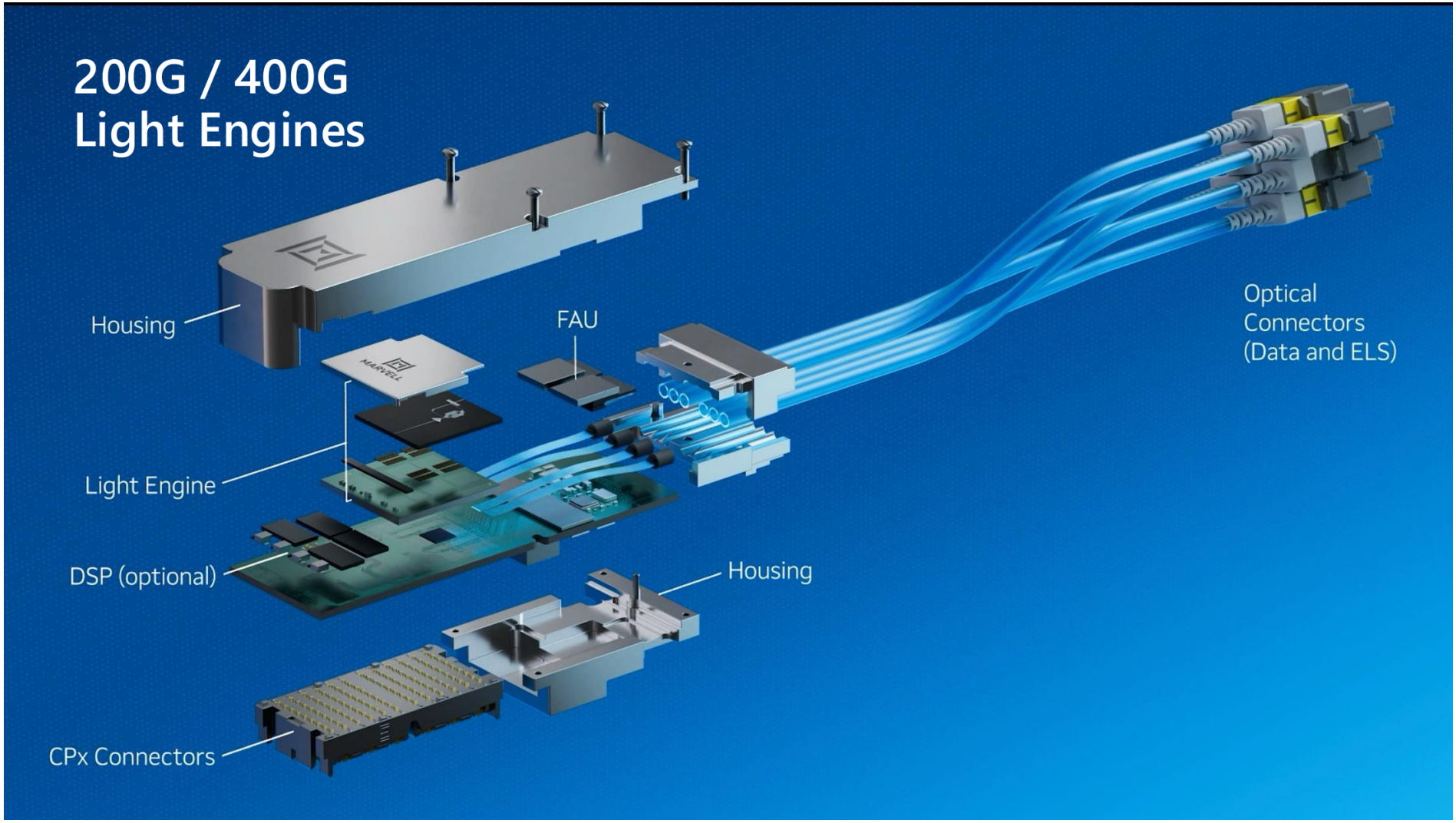
Coherent reception uses an optical reference and recovers phase as well as amplitude. Source noise, wavelength stability and the receiver/DSP design therefore matter alongside delivered optical power. Marvell's public Aquila brief specifies O-band fixed-wavelength lasers for its 800G/1.6T platform, with a 2-20km application range. It does not provide a universal linewidth acceptance threshold for every Marvell engine. Aquila's older 400G-per-lane 16-QAM implementation should not be presented as the detailed bill of materials for the newer 2nm demonstration. [LM2]

Map the source to the sold product. Fixed-wavelength O-band laser content belongs in CW or UHP chips according to the accepted SKU, and in ELS when Lumentum sells the complete external module. The separate narrow-linewidth assembly line primarily represents tunable C/L-band coherent products. A narrow-linewidth specification does not create another tunable-assembly sale. This distinction prevents counting the same source in both product pools. The Q4 2027-Q3 2028 pre-premium references are $1,010M CW, $1,035M UHP and $190M ELS; the separate narrow-linewidth assembly forecast is $761M. These are existing model allocations, not disclosed customer program revenue.

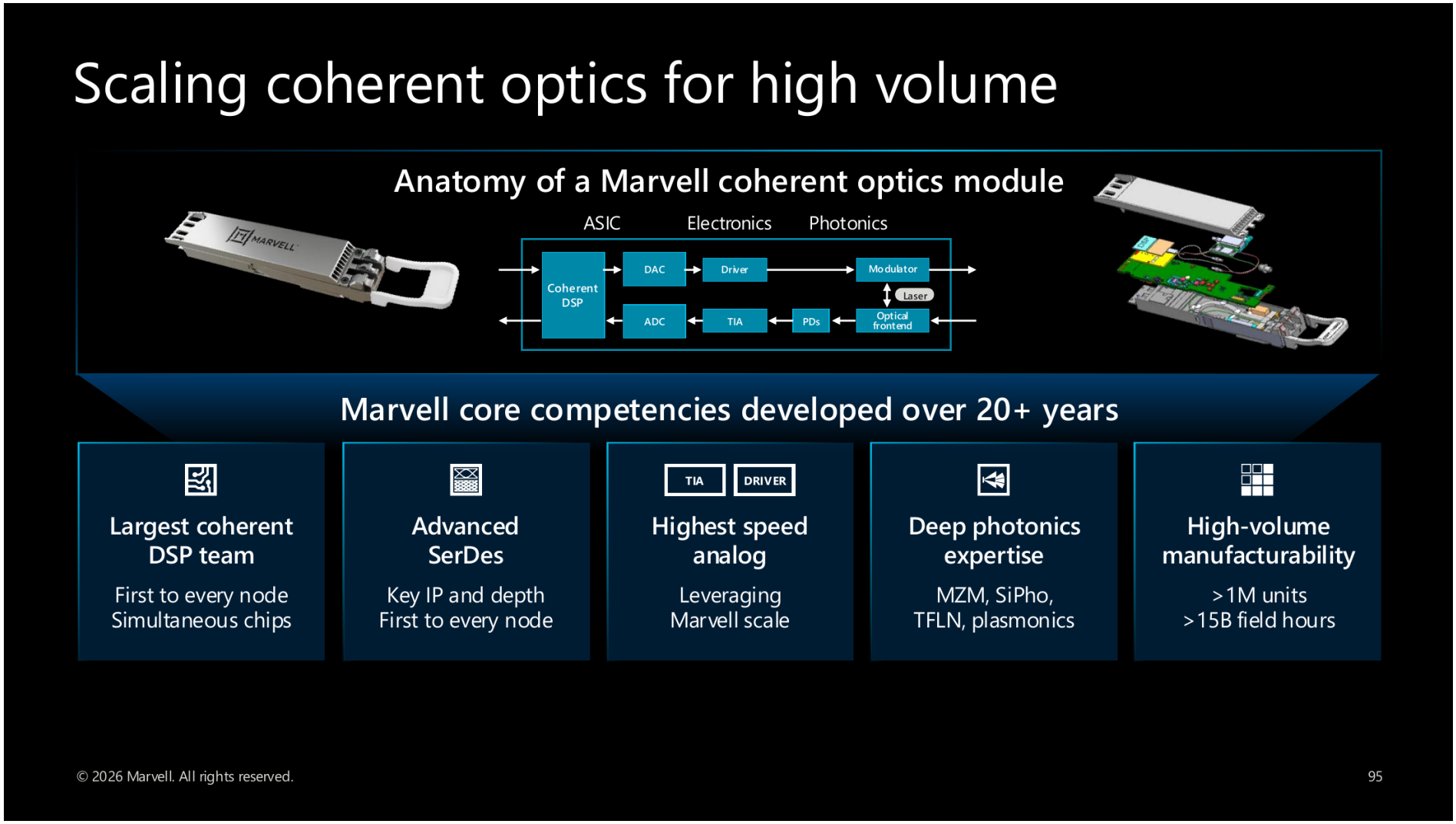
Lumentum's March OFC disclosure is relevant product evidence: its SHP source exceeds 800mW at 50C and is described with linewidth below 100kHz for direct-detect and coherent-lite applications. Those are issuer specifications, not a matched comparison proving superiority over every competing source or an announcement that Marvell selected that laser. Customer-relevant comparisons must preserve power at the fiber, temperature, drive current, linewidth convention, RIN measurement bandwidth, cooling, lifetime and manufacturing yield. [LM3]

Narrow linewidth is a performance property, not automatically the name of a revenue line. O-band fixed-wavelength demand belongs in CW, UHP or ELS according to the product sold. It should not also be added to the separate narrow-linewidth tunable-assembly path simply because coherent reception needs phase stability. Likewise, the older Aquila product brief is useful architecture evidence, not a specification sheet for every newer Marvell engine.

The economic chain is usable source power after coupling and splitting, then qualified sources per accepted engine, Lumentum's share and realized selling price. A more powerful source can serve more channels and reduce the number of seed devices. A seed-plus-amplifier design can shift content toward SOAs, packaging and test. The same optical engine cannot simultaneously create a full laser-chip invoice and a second external sale for that internally consumed chip. The existing split between externally sold UHP chips and ELS modules remains the correct accounting boundary.



Marvell Investor Day, PDF page 113. The light-engine assembly labels optical connections for data and ELS, plus an optional DSP. This establishes an external-light architecture, without identifying a supplier, laser count, source power or ASP. LITE chip and complete-ELS invoices are different scopes.



Marvell Investor Day, PDF page 95. A complete coherent module contains DSP, analog electronics, modulator, detector and laser functions. A qualified light-source supplier earns its own invoice, not the entire module price. The more-than-one-million units and field-hour claims are cumulative Marvell experience.

A real OCS relationship, with a separate commercial question

The March 16 Marvell-Lumentum announcement names a live interoperability demonstration between Lumentum's R300 optical circuit switch and Marvell's Ara, Aquila and COLORZ platforms. It establishes ecosystem compatibility. It does not identify every laser inside Marvell products or quantify additional Lumentum OCS orders. This March event predates the OFC commercial evidence already used in our September OCS model. [LM4]

Marvell's April 20 technical recap confirms that the live system combined R300 with different Ara, Aquila and COLORZ modules, with RELIANT telemetry. It supports system integration and compatibility across reach classes. The older January market forecast cited in that recap is not treated as a new October forecast or averaged with the later July OCS outlook. [LM6]

The active OCS case retains a 30% probability of the specified $10B calendar-2028 outcome. Its reference is $1.527B; the conditional increment is $8.473B, producing a $2.5419B weighted increment and $4.0689B active OCS revenue. The interoperability demonstration supports the route to market but does not independently substantiate the much larger endpoint or its 2028 timing. Raising that endpoint's probability solely because the same platform appears again would count the technical milestone twice.

Competition changes by link and by invoice

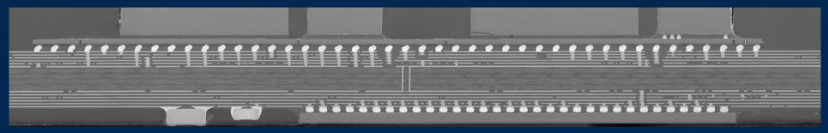
Marvell sells complete COLORZ modules as well as merchant DSPs. Its growth can support demand for laser components while competing with module suppliers for the finished invoice. This is a reason to track Lumentum's component and system margins separately. It does not imply that every integrated Marvell engine removes the external laser supplier.

Marvell also pursues microLED, microVCSEL and other photonic paths. Its March Mojo Vision agreement specifically concerns short-reach optical interconnects; page 67 labels the microLED route as board-level and laser-free. That is genuine substitution evidence for relevant sockets, not a replacement of long-reach, OCS, pump or all CPO laser demand. The development agreement and short-reach alternative predate the current model. The October deck supplies no newly earlier volume date or new displaced Lumentum allocation. [LM5]

Breaking the copper wall with photonics

3D integrated silicon photonics light engine

Electronics ▶
Photonics ▶
Substrate ▶
Integration on reverse side ▶



Pursuing every optical path, across every distance

High-speed, long-reach	Ultra-low power, high density	All-optical scale-up fabric	Board-level, laser-free	Short-reach, high-density	Ultra-fast, ultra-compact
MZM, TFLN	MRM	EAM	µLED	µVCSEL	Plasmonics, SiPho

More data movement drives more optical content at scale

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Marvell Investor Day, PDF page 67. Different links can use MRM, MZM/TFLN, EAM, microLED, microVCSEL or other photonics. The board-level microLED path is labeled laser-free. Substitution is architecture-specific; this is not a universal removal of InP demand.

Financial disposition under Rule 16b

The evidence was tested against the active model rather than excluded for lack of an announced award. The roadmap's 2027 engine timing overlaps the existing 2027 source ramp. Before the FCC and scarcity premiums, the Q4 2027-Q3 2028 UHP/ELS reference pool is $1.225B. The existing 55% probability applies to a bounded 5% premium on 50% of that pool, followed by a 50% overlap deduction. After weighting the 2027 portion at 70% for timing risk, this produces $14.43125M conditional incremental revenue and $7.937188M weighted revenue. Incremental COGS of 10% and operating expense of 5% leave $6.746609M weighted operating profit, or about $1.2912 per share at the retained 20x multiple and 104.5M shares. This contribution is already active.

Marvell's roadmap corroborates optical adoption; it does not separately improve the price-premium outcome beyond the existing management-supported scarcity case. The 55% probability therefore stays unchanged. The 30% OCS and 40% FCC cases also stay unchanged. The new deck receives a zero additional revenue, COGS, opex and OP increment because its demand/timing and technical-compatibility mechanisms are already represented, not because confirmation is required. It adds no new factory, independent customer wallet, royalty stream or laser price assumption.

Economic outcome	Selected probability	Existing contribution	Additional Marvell-deck economics	Disposition
Qualified UHP/ELS repricing	55%	Active-window revenue $7.937188M; OP $6.746609M; $1.2912/share	$0 additional revenue and OP	Existing 5% premium, 50% repriceable pool and 50% overlap already capture the bounded price outcome. Roadmap adoption is not independent evidence of a larger accepted premium.
OCS $10B CY2028 endpoint	30%	$2,541.9M weighted increment above $1,527M reference; $4,068.9M active CY2028 OCS	$0 additional revenue and OP	March/April interoperability predates the reference evidence and does not substantiate the larger endpoint or its timing.
FCC pricing scenario	40%	Active-window OP $16.3395M	$0 additional revenue and OP	No new policy, content rule or qualifying-product evidence is contained in the Marvell deck.

Two bounded probes make the no-increment decision testable. A 30%-weighted 5% extra qualified-UHP-output case with 50% overlap and 2% additional qualification expense would add about $6.004245M of active-window OP and $1.1491/share after ordinary production and operating costs. A revenue-neutral 10% CW-to-UHP dollar-mix shift at 30% probability, with 2% qualification cost, would add $6.246750M OP and $1.1956/share. These are diagnostic assumptions, not selected inputs. The source discloses no incremental good-unit capacity or allocation above the existing ramp, and linewidth alone does not justify replacing a CW margin with the much higher UHP assumption. Counting total industry coherent-module units as the new fixed-wavelength cohort would also mix O-band and ZR markets. The precise unresolved inputs are accepted source SKU, source count and power per engine, customer allocation, net incremental volume, realized ASP and manufacturing yield. Existing active pricing and OCS probabilities remain nonzero; no new source-specific profit stream is inferred from the diagram.

The next changes should be driven by genuinely different economics: qualified source output or net realized price above the retained path; a changed laser/ELS mix with measured assembly costs; commercial adoption ahead of the existing timing; or a specific short-reach substitution rate that changes the modeled product pool. The current model's aggressive UHP margins, fixed-customer commitments and competing architectures remain counter-evidence to an automatic probability increase.

Sources: LM1: Marvell ECOC demonstrations, September 20, 2026; LM2: Marvell Aquila product brief, revised December 2024; LM3: Lumentum OFC product disclosures, March 17, 2026; LM4: Marvell-Lumentum OCS demonstration, March 16, 2026; LM5: Marvell-Mojo Vision development agreement, March 12, 2026. LM6: Marvell technical recap of the live OCS demonstration, April 20, 2026. Original October 6 deck PDF pages 67, 84, 95, 111 and 113 are embedded intact above. Source deck SHA-256: 1e6bc0130675f91e1f5ec9ef2e27e83df7a70658d815244c2f2db4a07b52a340. Page references use one-based PDF numbering. All scoped scarcity,OCS andFCC amounts in this chapter reconcile to their retained LITE_Q_Model branches; the separate VCSEL economics are in the current ledger above.

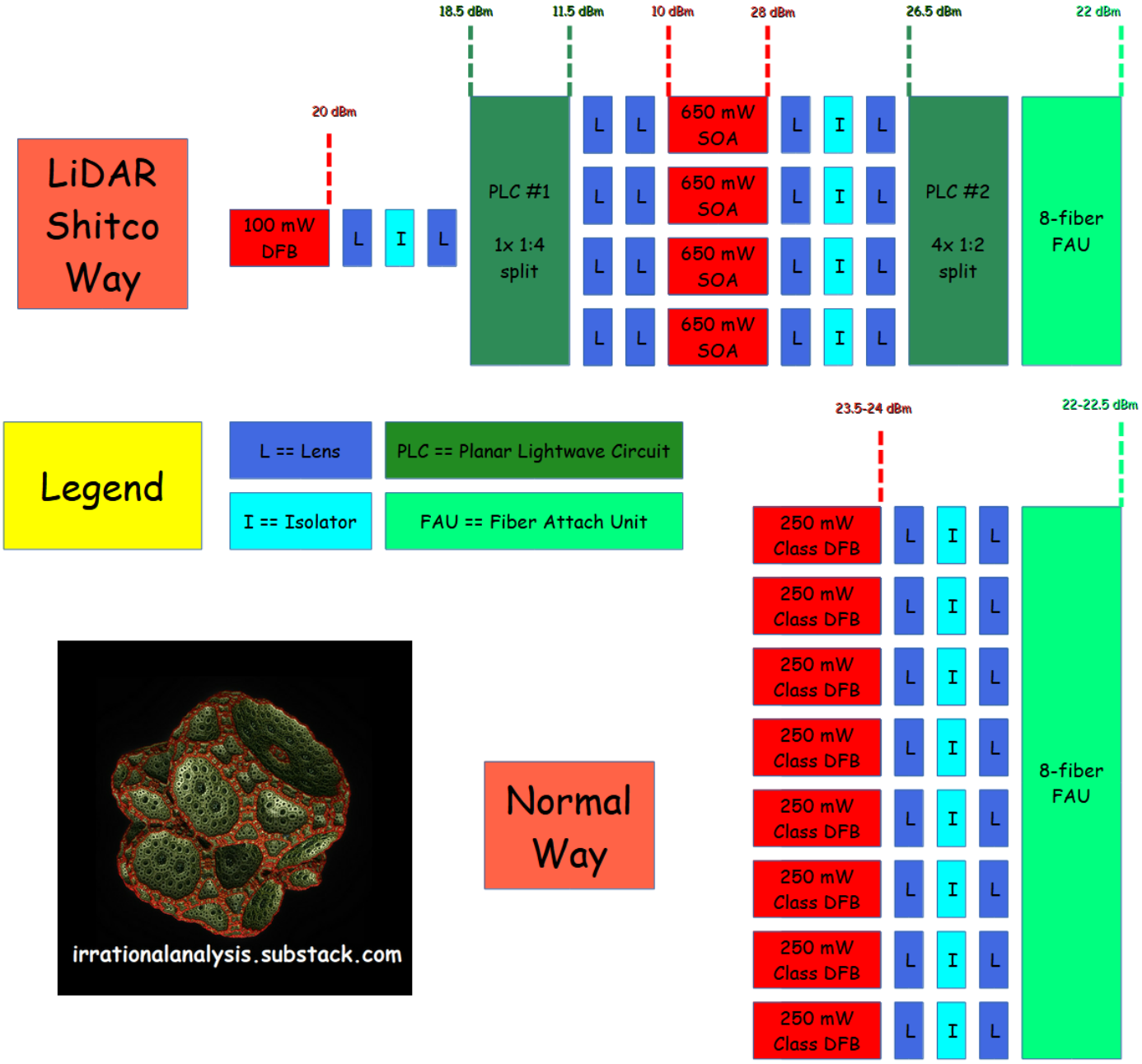
Qualified light: noise, power and assembly economics

The three July articles add the engineering background to Irrational Analysis's later earnings notes: July NPO/CPO Update, July 11, Practical Semiconductor Optical Amplifiers, July 18, and Stranded Indium-Phosphide, July 25. Their central mechanism is that a laser must supply usable light after noise, heat, splitting and coupling losses. Higher output by itself does not establish a better light source. Nor are these three articles independent confirmations of the author's August and September argument.

A distributed-feedback laser uses its grating and cavity to select the optical mode. Temperature, current and carrier distribution can change that operating state. The July article's low-power mode-hop graph is near 780 nm at 2 mW; it illustrates a mechanism, not Lumentum's 1310 nm high-power production yield. Its grating drawing is explicitly AI-generated. Both are explanatory aids rather than comparative vendor measurements. The author's approximately 0.1 nm/K drift and longer-cavity arguments must be applied to a specified device and operating envelope, not used as fixed universal manufacturing laws.

The SOA alternative clarifies the value of a strong seed laser without proving that only one architecture can work. A semiconductor optical amplifier can raise output, but also adds spontaneous-emission noise and can introduce gain saturation, feedback and wavelength interactions. An optical isolator can suppress returned light; it does not remove every noise contribution. Keeping one wavelength in each amplifier reduces interchannel four-wave mixing, while leaving intensity noise, thermal control and packaging to solve. The July 18 figures include cross-gain and cross-phase wavelength conversion as well as four-wave mixing; these are related nonlinear effects, not interchangeable tests.

The author's July 25 design targets eight fiber outputs at one wavelength, 1310 nm. It compares eight 250 mW-class DFBs with one 100 mW seed divided among four 650 mW SOAs, then split into eight outputs around 22 dBm each. This is a design calculation. It differs from July 18's eight-wavelength/eight-SOA example, so the lower device count is not a demonstrated improvement in the same product. The assumed seed and amplifier power must survive actual lens, isolator, splitter and fiber-coupling losses.



July 25 author design calculation: one wavelength delivered through eight fibers. This is not a production BOM or measured module.

Its cost workbook illustrates a plausible assembly disadvantage for the discrete SOA route, but the claimed 51% versus 17% module margins are not reported Lumentum or competitor results. The workbook assumes a $300 module selling price, 95% versus 85% assembly yield and specific component/alignment costs. The row labeled 12% contract-manufacturer gross margin is calculated as a 12% markup on cost. At a true 12% gross margin, the comparable illustrative downstream margins are approximately 50.2% and 16.1%. The listed BOM does not explicitly price the PLC splitters, fiber array, housing, control electronics, testing, warranty or NRE. Some may be bundled elsewhere, but the workbook does not establish that. The useful conclusion is that assembly and saleable yield can preserve differentiation, not that the exact margin is known.

	Normal Way	LiDAR Shitco Way	Comment
# DFB	8	1	
DFB Cost $	2.6	3.5	**DFB vendor gets margin from here. InP device model assumes sale external. This sheet assumes consign internal devices to contract manufacturer.**
# Lens	16	18	
Lens Cost $	1.5	2	Custom lenses for strange beamshape. Need custom over off-the shelf to make tight optical path budget.
# Isolator	8	5	
Isolator Cost $	2	2	
# Active Align	16	19	
Active Align Cost $	1.5	2	Tighter tolerances
# TEC	4	5	2 DFB per TEC possible
TEC Cost $	10	12	
# SOA	0	4	
SOA Cost $	0	10.2	**LiDAR shitco gets margin from here. InP device model assumes sale external. This sheet assumes consign internal devices to contract manufacturer.**
BOM COST $	124.8	188.3	
Yield Penalty %	95	85	
CM Final Cost $	131.4	221.5	
TFC/Fabrinet/Accelink Gross Margin %	12	12	
ASP to CM Customer $	147.2	248.1	
ASP to Hypersclaer $	300	300	
Laser or Lidar Shitco FINAL Gross Margin $	51	17	

Author cost/yield sensitivity. The 12% CM line is a markup on cost despite its gross-margin label; these are assumed costs, not issuer results.

The current model already prices this mechanism. “The ultra-high-power split” uses 85% chip and 55% module anchor margins, with reference peaks of 90% and 59%. “Qualified laser scarcity” adds a 55%-weighted 5% conditional premium on half of new/repriceable UHP/ELS sales, after a 50% overlap deduction, with incremental production/support costs. These July articles therefore receive full overlap with the existing economics. They add no factory output, customer allocation, accepted price, OCS demand or separate royalty stream. The current 30% OCS and 40% FCC cases remain separate and unchanged. The Q4 2027 price target remains $1,143.874580, using 20× forward-four-quarter operating profit and 104.5M shares; analytical Base 80 and Portfolio Excel Base 99 remain intact.

The productive next comparison is a matched operating point: fiber-delivered power, temperature, drive current, linewidth definition and observation time, RIN bandwidth, module electrical/cooling power, production-lot distribution and customer yield. Lumentum’s already reviewed 50 C curves and separate SHP development are stronger evidence than an unmatched headline power comparison. An alternative amplifier route can both sustain today’s packaging premium and eventually compete with it; neither effect creates an independent revenue addition here.

Policy engineering is separate from policy economics. The August 5 article argues from CMIS authentication to a broad security conclusion. OIF CMIS 5.4, section 7.3.3 makes module authentication optional; its later protocol description does not mean all deployed modules are authenticated, payload-encrypted or free of firmware risk. Nor does it prove that a particular module is compromised. This technical correction does not create a second regulatory event. The existing October 1 40% policy outcome, filtered product exposure and recognition timing remain controlling; the older August discussion is fully overlapping evidence, with zero new revenue, COGS, opex or OP.

Review scope: complete saved July 11, July 18 and July 25 articles and all 43 image elements were accounted for. One locally saved SOA figure was a placeholder; its original illustration was recovered separately from the cited 2017 research paper and retained in the audit. Original submitted files were preserved. Generic teaching diagrams, the author's spreadsheet assumptions and issuer measurements have distinct evidentiary status.

Array and amplifier measurements: scope matters

The NVIDIA/Lumentum ECTC 2026 array provides a concrete example of the distinction already made between a laser die and a delivered light-source module. Nine DFB-SOA channels share optical packaging and a TEC; the reported module efficiency is 7.2%, including passive losses. The paper’s roughly 9-10% model targets refer to specified link-energy budgets and exclude TEC in that subsection, so they must not be compared directly with a 400 mW bare-device PCE curve. Figure 5 also reports channel-dependent RIN and identifies further noise improvement for some 32 Gb/s/wavelength DWDM applications. It does not establish universal failure of Lumentum’s UHP portfolio. Shared alignment may reduce assembly operations, while common temperature control, wavelength binning and array yield impose different costs. No production-yield or COGS distribution is supplied. [IA08 images 024-030, 032-033; P01/P02]

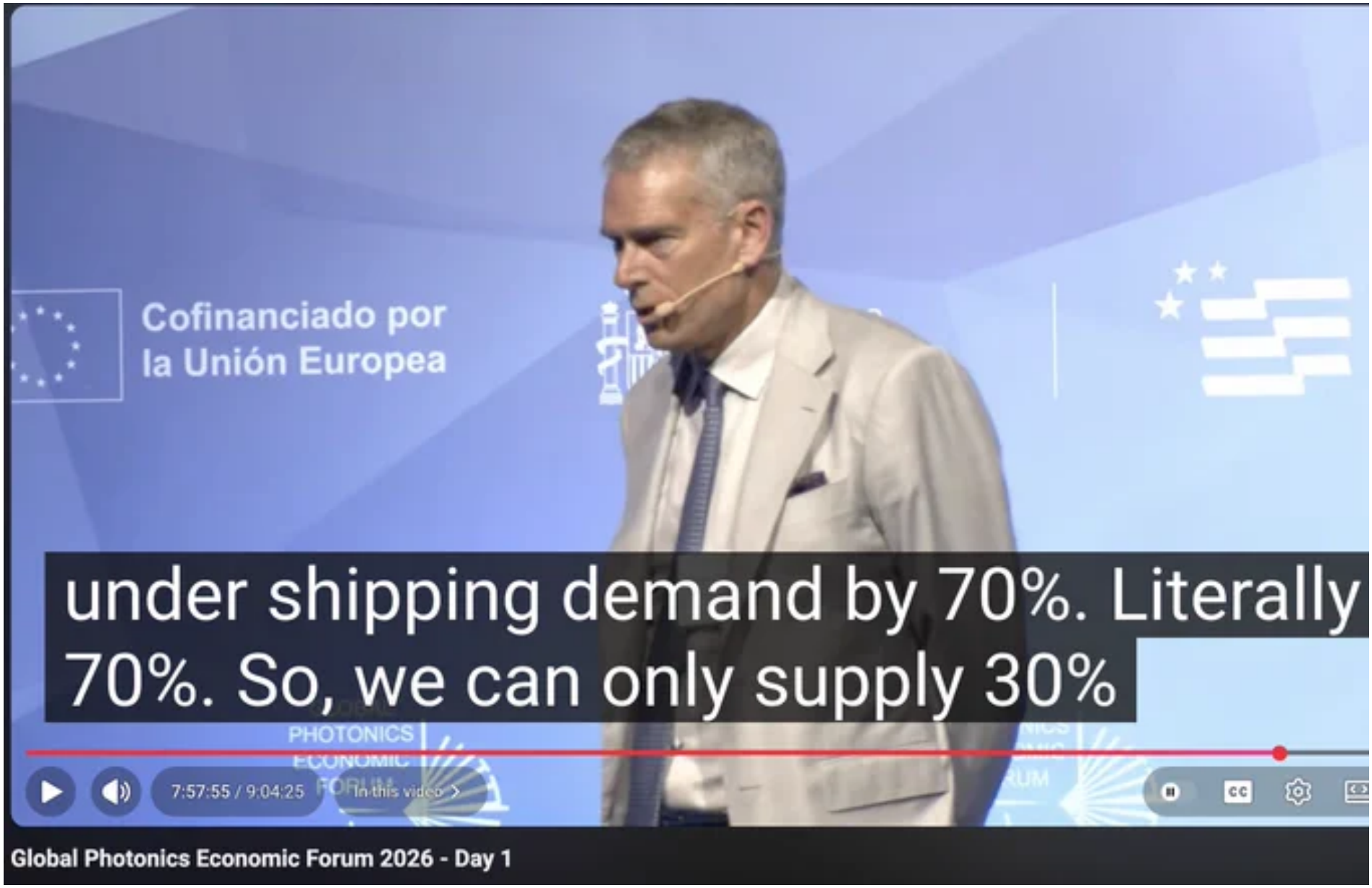
This evidence adds measurable qualification and cost questions to the existing scarcity case. The 55% bounded repricing branch already captures the same qualified-light scarcity, the 40% policy branch uses a filtered product pool, and the 30% OCS branch concerns a separate system product. The paper supplies no additional contracted capacity, accepted price or incremental shipment schedule. Retain those forecasts and use the figures to test module loss, cooling, noise, lot variation and customer acceptance.

P01: Design and Packaging of a DWDM CW-DFB Laser Array for Co-Integrated Optical Interconnects (2026-05-26 conference; IEEE added 2026-06-17). P02: NVIDIA research record of the DWDM laser-array paper (2026-05-26). Detailed paper figures were inspected in the submitted article; full IEEE papers were not independently recovered. These measurements refine acceptance tests, not customer allocation or commercial yield.

Qualified laser scarcity: primary evidence and active economics

Lumentum CEO Michael Hurlston's September 24 appearance at Optica's Global Photonics Economic Forum supports a constrained 2027 supply outlook. The October 2 review of captions on Optica's official Day 1 recording located the question and answer: in the context of expanding indium-phosphide manufacturing and the arrival of CPO/NPO, he estimated that Lumentum could supply roughly **30% of customers' requests in 2027**, implying approximately **70% undersupply**. He expressed hope for some balance around **2029-2030**, not a committed recovery date.

Optica event and recording chain | Official agenda | Capacity question at 7:57:15 | Numerical answer at 7:57:46 | Balance expectation at 7:58:33



Optica video still preserved in Serenity's October 2 post. The recording's captions support the customer-request shortage statement; the image is a secondary reproduction of the primary appearance, not a quantified order book.

The denominator is **requests made to Lumentum**, not measured worldwide demand. The passage gives no audited unit count, dollar value, product mix, ASP, margin, or adjustment for customers requesting supply from several vendors. It does not establish an EML-only, UHP-only or company-wide revenue shortfall. The evidence review read the primary video's timestamped captions; the audio was not separately transcribed. Speech-recognition errors elsewhere in those captions warrant short quotations and careful scope labels. Serenity's original relay is the same statement, not an independent demand datapoint.

Probability: 55% for the bounded incremental economic outcome below. The primary management source supports the shortage statement's provenance. It does not by itself increase the probability of our separate 5% price-premium hypothesis: unreported product mix, fixed customer contracts, already-high modeled margins and overlapping scarcity economics still constrain realization. The 55% weight is a subjective forecast judgment, not the probability that the CEO spoke or a measured frequency. The 50% overlap deduction remains part of the active bridge. The October 11 review also applies a separate 30% conditional timing-delay risk to the existing 2027 premium only: this is a delay of an already defined cohort, not a second independent demand event.

Driver	Selected assumption	Reason
Conditional additional premium	5%	Bounded price negotiation outcome, not management pricing guidance
Exposed UHP / ELS revenue	50%	Proxy for new or repriceable sales; fixed commitments insulated
Overlap deduction	50%	Existing scarcity, high product margins and the FCC contribution already capture much of this signal
Active weighted premium at full phase	0.6875%	55% × 5% × 50% × (1 - 50%)
Cost of incremental price revenue	10% COGS + 5% operating expense	Allows support, qualification and fulfillment costs; original production costs retained
Phase	17.5% Q1 2027; 35% Q2; 52.5% Q3; 70% Q4 2027; 100% in 2028-2029; 50% in 2030	Analyst timing and fade, including 30% conditional risk of delayed 2027 repricing; no catch-up

Reference UHP/ELS revenue is measured **before** the saved FCC premium. The two increments are additive; neither reprices the other. There is no 3.33× revenue multiplier, extra unit shipment, new factory, or additional OCS outcome. Already-scarce CW and EML sales are excluded from this narrow transmission. Product margins remain below 100%; incremental gross profit is 90% and incremental operating profit is 85% of this price revenue.

Quarter	Revenue increment USD M	COGS	Opex	OP increment
Q3 2026	0.000	0.000	0.000	0.000
Q4 2026	0.000	0.000	0.000	0.000
Q1 2027	0.126	0.013	0.006	0.107
Q2 2027	0.349	0.035	0.017	0.297
Q3 2027	0.686	0.069	0.034	0.583
Q4 2027	1.131	0.113	0.057	0.961

Quarter	Revenue increment USD M	COGS	Opex	OP increment
Q1 2028	1.959	0.196	0.098	1.665
Q2 2028	2.269	0.227	0.113	1.928
Q3 2028	2.578	0.258	0.129	2.191
Q4 2028	2.853	0.285	0.143	2.425
Q1 2029	3.094	0.309	0.155	2.630
Q2 2029	3.334	0.333	0.167	2.834
Q3 2029	3.541	0.354	0.177	3.010
Q4 2029	3.747	0.375	0.187	3.185
Q1 2030	1.959	0.196	0.098	1.665
Q2 2030	2.035	0.204	0.102	1.730
Q3 2030	2.111	0.211	0.106	1.794
Q4 2030	2.186	0.219	0.109	1.858

The scarcity bridge contributes 6.747 million dollars of operating profit in the Q4 2027 forward window. At 20× operating profit and 104.5 million shares, its contribution is 1.2912 dollars per share. Analytical Base is 80: demonstrated scarcity already supports Core 18, Macro 10 and Catalysts 5, while concentration and 2028 capacity risks remain.

A second digest post places a broad 3.2T ramp in 2029. This is relevant counter-evidence to the saved early-sourcing FCC case, whose full intensity begins Q2 2028. The existing 40% FCC case is retained as an explicitly early-sourcing sensitivity, not relabeled as an official date. Original MS scope/timing and customer contract repricing remain high-priority evidence requests. Jeffrey Pu’s attributed 12%/26% CPO revenue mix is not added: its product denominator overlaps this UHP/ELS path and cannot currently be reconciled to a distinct revenue pool.

Pricing and licensing commentary: one overlapping signal

Fabian's October 2 post suggests licensing ultra-high-power CW laser IP or charging a larger premium. His reported brief CTO discussion concerns reverse-engineering risk from revealing laser cross-sections; it is not a disclosed licensing agreement, royalty schedule or price increase. Zephyr's pricing comment likewise argues that management could extract more ASP without specifying an accepted customer price. These opinions concern the same scarcity monetization already represented by the **55% bounded premium case**. Their separate incremental revenue, operating profit and price target contribution is **zero because of overlap**, not because social evidence requires official confirmation.

The quoted optics commentary explicitly lacks confirmation of Coherent's 400 mW qualification, particular technical problems or ability to meet its December-quarter volume objective. Absence of that information is **not a confirmed qualification failure**. No competitor-loss revenue, exclusive-supplier assumption or additional licensing revenue enters this model. Revisit a separate royalty bridge when license scope, commercial terms and cannibalization of product sales can be bounded;

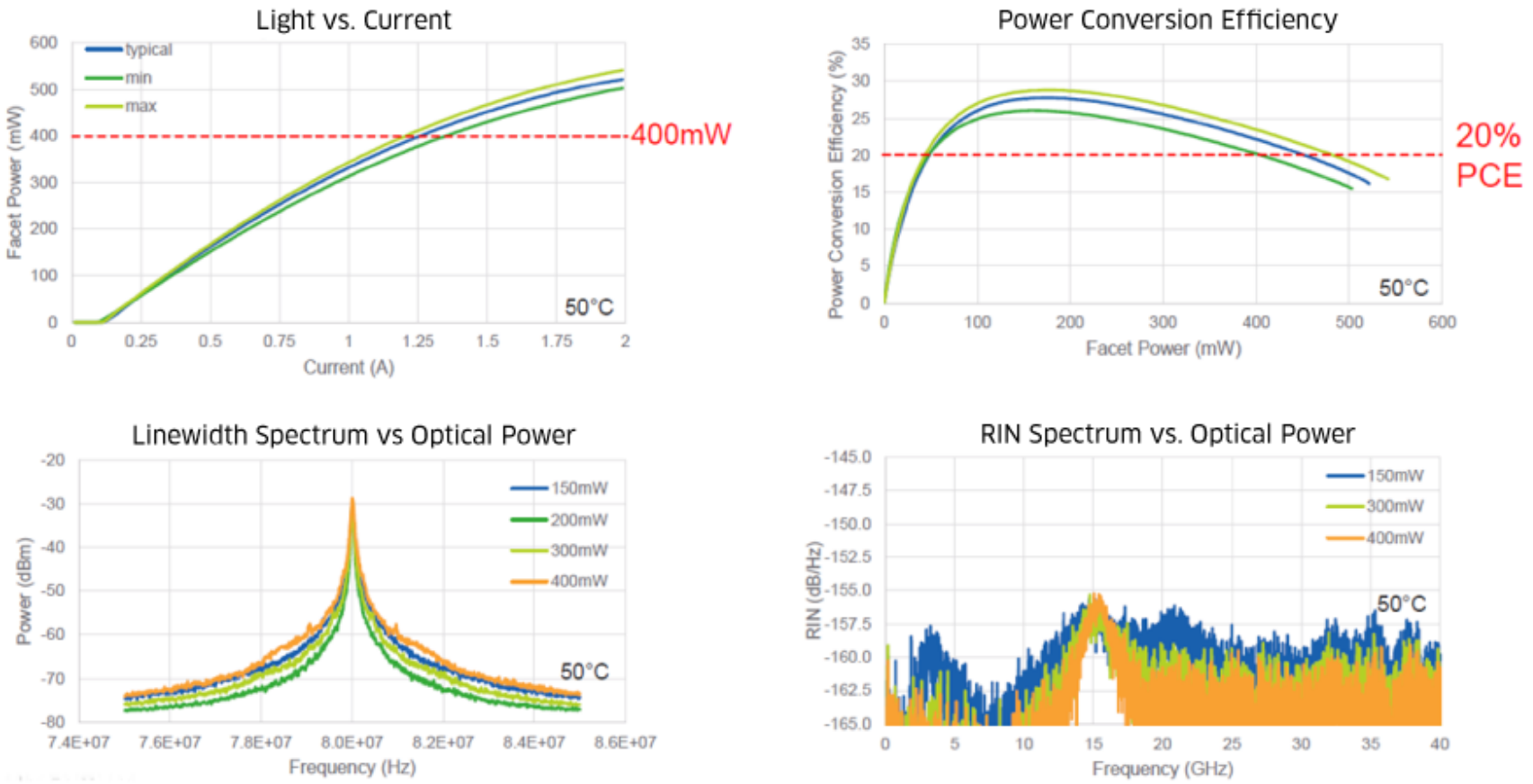
revise the existing premium only when customer price, product mix or cost evidence changes the defined outcome.

Engineering evidence: why qualified light can command a premium

Lumentum's technical differentiation is credible. The strongest evidence is its operating envelope across power, efficiency, noise and temperature, combined with customer yield and qualification economics. The public material does not establish a universal ranking across every laser application, nor a universal failure by Coherent. The relevant question for this model is whether Lumentum can retain superior economics on qualified high-power lasers while demand exceeds deliverable capacity.

The initial three earnings/laser articles by **Irrational Analysis** are: Lumentum earnings, August 11, Coherent earnings, August 14, and September 21 follow-up. These are one analyst's connected argument. The June 13 coherent-lite background article is reviewed separately below; none of the four is an independent confirmation of another. The three earnings/laser bodies include 57 image occurrences; the June background body adds 41, all reviewed. The June page also has two visible comments and 18 additional comments absent from the saved file. Nine additional comments and two linked attachments were not present in the saved webpages. Primary issuer figures and the relevant standards were separately checked. The author reports firsthand Lumentum measurements; those remain attributed engineering testimony, with no supplied raw dataset enabling replication.

Four measurements, four different investment implications



Lumentum primary Figure 3, June 25, 2026. All panels are labeled 50 C. Power, PCE, optical spectrum and RIN describe different quantities; the figure is not a matched Coherent phase-noise test.

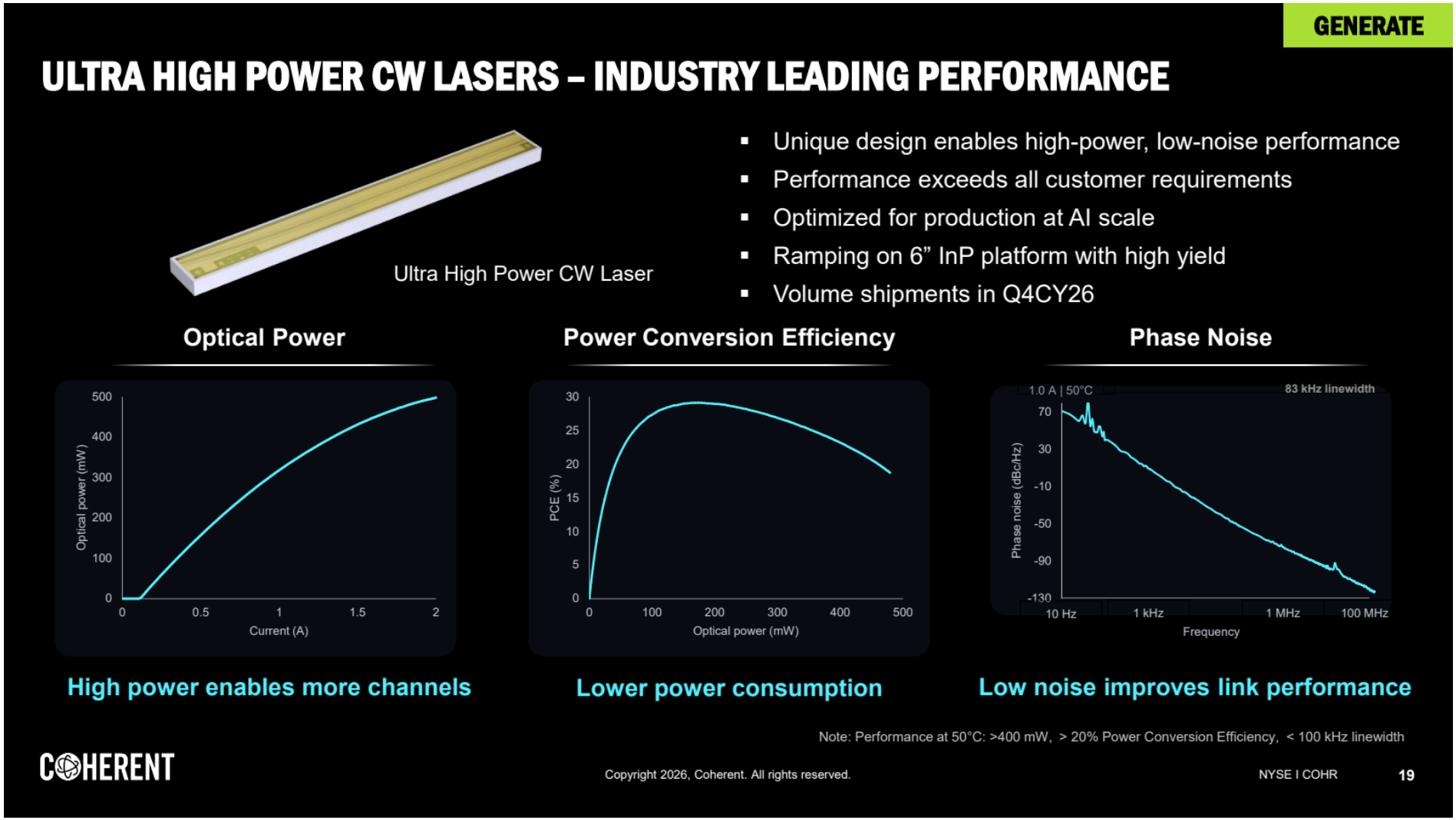
Lumentum primary figure and capacity discussion. The device demonstrates approximately 400 mW around 1.3 A and above 20% power-conversion efficiency at that optical power. Efficiency is optical output divided by electrical input. Package coupling losses and cooling power must still be included when estimating an external laser module's wall-plug efficiency.

Engineering feature	Why it matters financially	What the evidence does not establish
High delivered optical power	Can support more channels per laser, subject to splitting and coupling losses	Device-facet power is not automatically fiber-delivered power

Engineering feature	Why it matters financially	What the evidence does not establish
Efficient operation at elevated temperature	Can reduce electrical and cooling burden in a customer's system	Peak device PCE is not the efficiency of a cooled ELS module
Low intensity noise and suitable linewidth	Can protect link margin and avoid rework or failed qualification	RIN, optical spectrum and phase-noise plots are not interchangeable
Stable operation across current, temperature and production lots	May improve customer manufacturing yield and sustain pricing	A single typical curve does not demonstrate a production distribution

The source's distinction between intrinsic and effective linewidth is relevant. Low-frequency noise can make a practical linewidth larger than the white-noise contribution alone suggests. Observation time, integration bounds, calibration and the convention used to convert phase noise to frequency-noise density matter. The source supplies a qualitative comparison, not a reproducible matched-device experiment.

Competition requires the same operating point



Coherent primary PhotonLink presentation, September 21, slide 19. The phase trace is labeled 1.0 A, 50 C and 83 kHz; the neighboring power curve places 1.0 A near 300 mW. The separate >400 mW capability is not an explicitly matched linewidth measurement.

Coherent original presentation, slide 19. Coherent claims customer requirements are met, high-yield six-inch production and Q4 calendar-2026 volume shipments. Those are company statements. The phase trace gives a reason to examine low-frequency noise more closely, while the published method is insufficient to declare every CPO/NPO design unqualified.

OCI v1.0, Table 2-6 sets a 1 MHz maximum laser linewidth, RIN of -144 dB/Hz and specified wavelength bands. It does not prescribe a beta-separation calculation or observation time. OIF ELSFP section 3.4 leaves application-specific optical requirements to the receiving engine/CPO specification. Applying an undocumented effective-linewidth method as a universal CPO pass/fail standard overstates the evidence.

The August article's -155 versus -135 dB/Hz RIN comparison is hypothetical. It supplies no Coherent -135 measurement. A 20 dB power-density difference would imply 100 times the variance, or 10 times the RMS fluctuation over the same bandwidth, not 100 times the amplitude. The September Coherent chart measures phase noise, a separate quantity.

Probability and incremental economics

Rule 16b field	Active treatment
Economic outcome and probability	Existing 55% probability of the bounded 5% UHP/ELS repricing outcome
Conditional pool and cost bridge	50% new/repriceable UHP/ELS sales; 50% original overlap deduction; 10% incremental COGS and 5% incremental opex
Additional overlap for these articles	100%; the August BOM and qualification argument already underpin product margins, and the September competitive concern reinforces the same scarcity outcome
Additional revenue, OP and target contribution	Zero from this review; the existing scarcity bridge still contributes $6.747M F4Q OP and $1.2912/share at Q4 2027
Deliverable unit constraint	No new fab output or customer allocation is established. Missed Coherent shipments do not become automatic Lumentum revenue
Analytical Base	80; Core 18 already captures qualification and pricing differentiation, with capacity and competitive risks retained

This is a quantified overlap decision, not a requirement to wait for an official announcement. The present revenue forecast already assumes very strong UHP margins and high utilization. Another premium on the same products would count the same mechanism twice. Material customer price terms, a larger qualified operating window, production-yield evidence or additional shippable capacity can change the selected inputs.

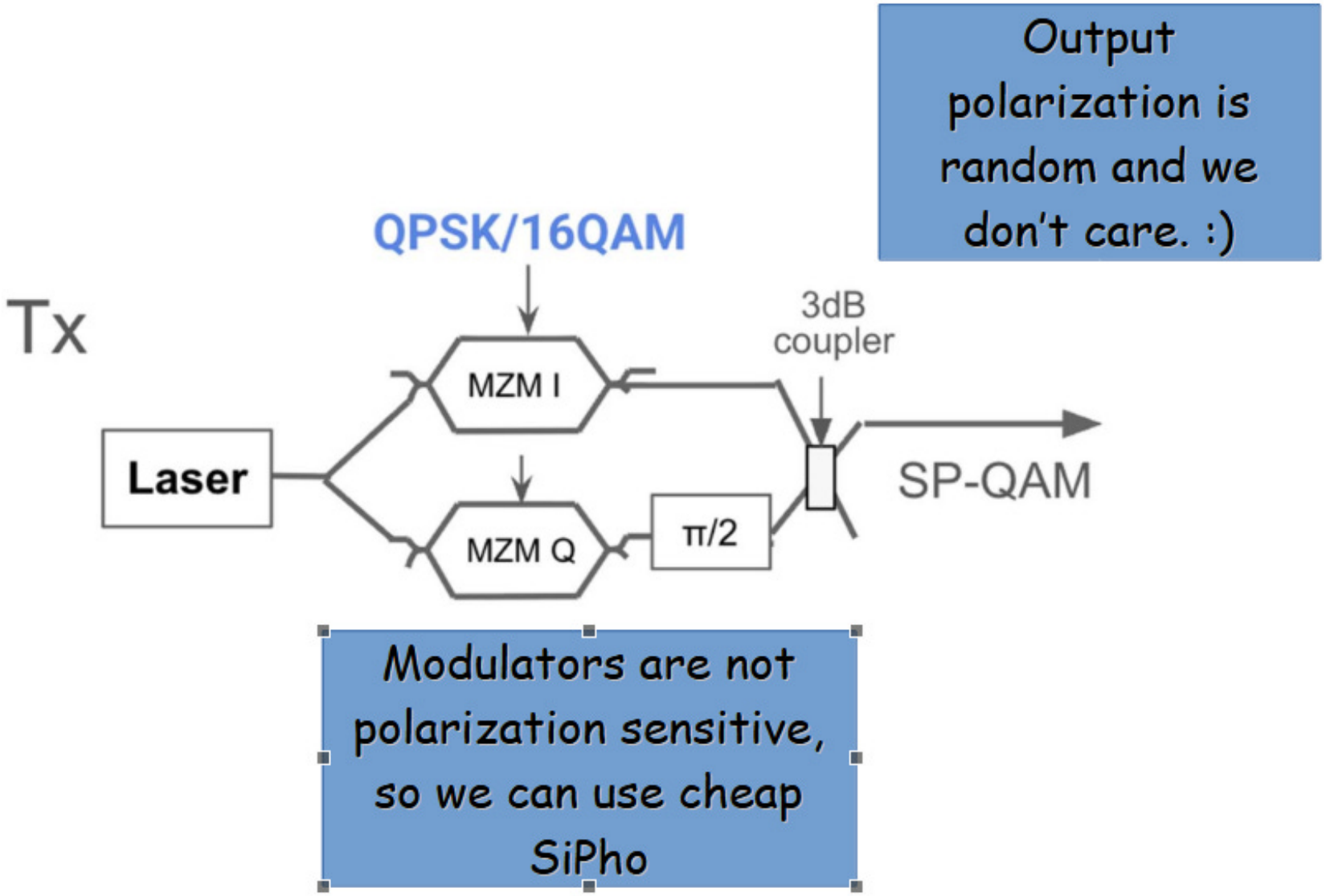
The most useful next comparison is production-lot data at matched optical power, temperature and drive current, with stated phase-noise bandwidth and observation time, followed by customer acceptance and shipment evidence. Lumentum's separate March 2026 SHP demonstration above 800 mW at 50 C is a further development milestone, not a substitute for those tests or a relabeling of the 350-400 mW product.

First principles: architecture decides who captures the value

The fourth supplied article, Coherent Lite Transceivers + AMD CPU Opportunity, was published **June 13, 2026**. It is technical background predating the August earnings notes and September follow-up. **Coherent-lite names a simplified coherent optical architecture; it is neither Coherent's company name nor Lumentum's ticker.** Its source models are the same author's evolving estimates, not independent confirmations of the later earnings articles.

The useful question is which functions need scarce InP, which can move to silicon photonics, and what the customer pays for a qualified optical link. Higher optical demand does not force every design to consume the same InP area, laser power or external components.

A hybrid design can reward different suppliers at once



June 13 article: author-annotated coherent-lite transmitter diagram attributed to a Google paper. The blue notes are the analyst's interpretation. Silicon photonics remains polarization-sensitive internally; this drawing does not prove every modulator is polarization-insensitive or identify a customer-selected supplier.

The article proposes a DFB laser plus a silicon-photonics transmitter, with a potentially monolithic InP receiver and separate DSP/driver/TIA functions. This is a plausible design route, explicitly presented by the author as a best guess. It is not a confirmed Google bill of materials. The transmitter can manage a known guided polarization while the receiver handles the polarization arriving through the fiber; these are different design problems.

Polarization folding can recover received signal components in a form compatible with a simpler coherent receiver, at the expense of signal-processing requirements. Google's polarization-folding patent supports the architecture's feasibility; it does not establish deployment volume, product price or a supplier award. Nor does one architecture prove that silicon photonics must win every transmitter or InP every receiver.

There is an important counterpoint to the article's dismissal of its middle architecture. Google's OFC 2025 tutorial, page 50, describes joint transmitter/receiver folding as recovering 3 dB of link budget by using an otherwise discarded output. It trades extra structure for optical-power benefit; it is not simply an inferior design. The related February 2026 JLT paper also identifies reduced phase-noise tolerance from partial-response signaling, while its shown sensitivity simulations neglect laser phase noise. That supports studying the noise problem, not a named-vendor pass/fail verdict. The tutorial is from 2025, correcting the supplied article's OFC 2026 attribution.

For Coherent, a possible integrated receiver opportunity is a positive counterweight to concerns about one high-power laser. Its PhotonLink portfolio spans generation, shaping, routing and detection. For Lumentum, a shift to silicon modulators does not eliminate the need for a qualified light source, but different architectures can change required laser power, supplier content and scarcity intensity.

Noise requirements belong to a particular link

Coherent detection uses a local-oscillator laser and recovers phase as well as amplitude. Phase noise therefore matters directly to carrier recovery. Balanced detection can suppress common intensity noise; it does not simply cancel every phase-noise contribution. IMDD, coherent modulation and CPO/NPO packaging describe different aspects of the system and cannot be treated as interchangeable specifications.

The June article's sub-0.2 MHz linewidth discussion is coherent-system context, not a universal acceptance threshold for all 400 mW CPO/NPO lasers. The applicable modulation, receiver, DSP, optical power, temperature and measurement definition must be matched. Likewise, comparing 168 and 224 without distinguishing baud rate, bits per symbol and implementation bandwidth does not by itself certify the proposed design.

Good packaged devices matter more than wafer area

Product	100 mW CW	400 mW CW (0.5-1 MHz linewidth) *competent vendor	400 mW CW (0.5-1 MHz linewidth) *loser vendor	200G EML	Coherent lite Rx
width (mm)	0.25	0.5	0.5	0.25	2
length (mm)	1	3	5	1.5	5
area (mm^2)	0.25	1.5	2.5	0.375	10
wafer size (in)	4	4	4	4	6
dice per wafer	25888	4308	2526	17200	1513
Yield %	50	50	30	30	20
good dice	12944	2154	757.8	5160	302.6
Package yield %	50	50	50	50	80
sellable dice	6472	1077	378.9	2580	242.08
pkg cost $	500	500	500	500	1000
wafer cost $	500	500	500	500	700
process cost $	4000	4000	4000	4000	5000
COGS per dice $	0.8	4.6	13.2	1.9	27.7



June 13 illustrative InP intensity and cost model. Die dimensions, wafer maps, die yield, packaging yield and cost inputs are author assumptions. Its vendor labels are rhetorical; no named-company production data establish these yield differences.

June illustrative case	400 mW case A	400 mW case B	Coherent-lite receiver
Assumed die area	1.5 mm2	2.5 mm2	10 mm2
Assumed wafer diameter	4 inches	4 inches	6 inches
Assumed die yield	50%	30%	20%
Assumed packaging yield	50%	50%	80%
Modeled saleable devices per wafer	1,077	378.9	242.08
Modeled cost per saleable device	$4.6	$13.2	$27.7

The two laser examples illustrate roughly **2.9 times the unit cost** from combined assumed die-size and yield differences. This explains why qualification and manufacturing consistency can matter economically. It does not measure Lumentum's or Coherent's actual cost or prove a 50% universal InP yield. Six-inch versus

four-inch diameter provides 2.25 times geometric area before edge losses and yield; that is different from the fourfold area comparison between six and three inches.

The June 400 mW example assumes $500 wafer cost and approximately $4.6 total unit cost. The later August example uses $2,000 wafer cost and approximately $6 unit cost. The source assumptions changed. Neither is an audited BOM, so the two estimates must not be averaged or treated as independent support for an exact product margin.

What changes in the active investment interpretation

This background strengthens product-level analysis and the need to test saleable yield. It also supplies counter-evidence to a blanket negative view of Coherent: the author himself identifies potential receiver benefits. His Marvell, Tower, Semtech, Nokia and MACOM supplier outcomes depend on the proposed architecture and socket allocation; they are research leads, not established design wins.

No separate revenue pool is added for the June coherent-lite discussion. Its general optical-migration mechanism is already represented in the relevant source-demand growth, with **100% overlap assigned to repeating that mechanism**. This does not establish that every future customer wallet is fully mapped. Fixed O-band source content must be assigned by sold SKU to CW/UHP/ELS, rather than automatically added to C/L-band tunable assemblies. The new Marvell chapter makes that distinction and tests bounded output and mix alternatives. The June article supplies no independent unit-volume schedule or product-specific ASP/cost increment above the reference path. The zero addition follows that economic scope and overlap, rather than a requirement for official confirmation. Customer allocation, delivered source content and genuinely incremental economics remain explicit tests.

Lumentum's existing **55%** bounded UHP/ELS repricing probability, product margins and target path remain unchanged. The background does not justify another scarcity premium or a broad claim that Lumentum dominates every coherent-optics component.

The AMD CPU appendix and dated trading-position screenshots were reviewed for completeness. They do not provide incremental evidence for either laser model and are not used in these valuations.

Analytical Base and scored conviction

Category	Score	Evidence and limit
Core Thesis	18/20	Silicon cannot emit light. Roughly 60% of global EML supply from two Japanese fabs. Still shipping behind EML demand with no change in the imbalance. On CW, management reports being able to price up because customers see materially better transceiver yields from a narrow spec distribution, which is pricing power demonstrated rather than asserted. Pump lasers at 70-80% share and effectively sold out. Not a monopoly: Coherent and Sumitomo are real.
Cycle Stage	8/10	Revenue +109% YoY with operating margin +2,160bps YoY. Guided to another record. On high-power lasers management says it is "way behind" and "very much further behind" because demand accelerated, not because execution slipped. Mid-cycle with the CPO and NPO S-curve not yet started.
TAM Expansion	9/10	Ford lens. Phase 1 800G and 1.6T pluggables at 100%. Phase 2 NPO, ELS modules, OCS and scale-across pump lasers at 50%, all with dated commitments. Phase 3 CPO scale-up and in-tray OCS at 20%. NPO is described as completely additive and significantly increasing the optical TAM, and the first ELS module order is in hand.
Catalysts	5/5	FQ1 FY2027 report. First triple-digit OCS quarter guided now. First triple-digit ultra-high-power laser quarter in FQ3 FY2027. ELS deliveries 2H calendar 2027. Greensboro first revenue early 2028. New OFC targets.
Macro	10/10	DB1. Copper is at its practical limit as clusters scale. One hyperscaler's capacity between two AI sites is double the total global backbone it built over the last decade.
Alpha	2/5	Heavily covered, 24 analysts. The edge here is vector decomposition, not discovery.
Risks	-4	Scored individually below.
Total	48/60 = 80/100	High Conviction; price-independent analytical thesis score

Artifact analytical Base: 80/100. Portfolio Excel Base: 99/100. The portfolio preserves the October 8 owner override; analytical scoring is separate.

Qualified laser scarcity and pricing power already support Core 18, Macro 10 and dated catalysts 5. The policy premium is modeled directly in operating economics. It does not establish additional physics, a new committed customer, a separate TAM or an informational edge deserving another point. The category total is 48/60 = 80.

Base measures thesis quality. The 40% policy probability weights only the incremental operating economics, once; neither Base nor the policy weight is applied again to the final price target. Scores use integer category points on the /60 framework lattice.

Policy breadcrumb and economic hypothesis

Serenity's October 1 post attributes Washington meetings to Morgan Stanley and reports a possible 3.2T-focused restriction, with a potential 65% U.S.-value route for Chinese modules. The source expects laser pricing to matter more than broad module-share redistribution. Treat this as an attributed policy scenario; the original MS note and final FCC language establishing these exact terms were not recovered. The separate September 25 Senate proposal concerns sensitive federal systems and a five-year transition. It is not confirmation of these FCC terms.

Serenity policy post | Foundry follow-up | Separate Senate proposal

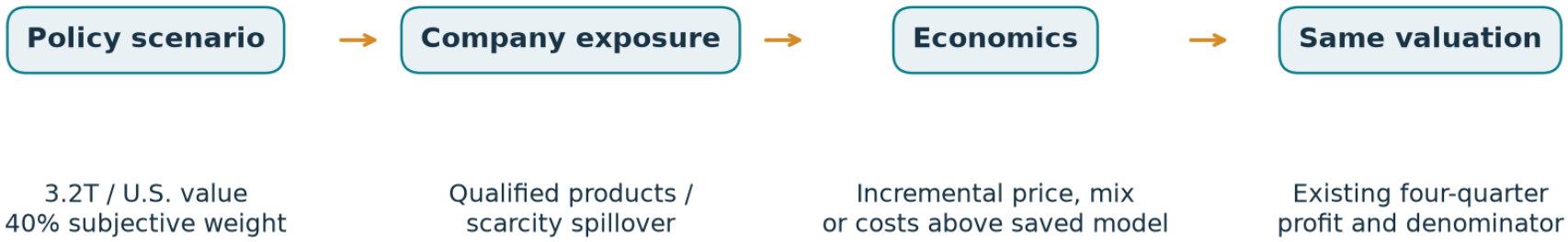


Figure 2. Analytical transmission of the attributed policy scenario. This is a model diagram, not an official regulatory diagram or customer award.

Assigned common policy-scenario probability: 40%. Subjective initial weight above the single-source 25-35% band because Serenity has the owner's highest weighting and attributes Washington meetings to Morgan Stanley. The exact future rule, timing and economic effect remain uncertain. Serenity and Morgan Stanley are one source chain, not two independent confirmations.

Incremental 5% premium only on a subset of CW/UHP/ELS external sales, above already modeled scarcity pricing. EML's Japanese supply, OCS, Cloud Light and the Greensboro capacity headline are not separately added. Fractional exposure avoids treating all product sales as qualifying U.S. content.

Eligible fractions are bounded scenario assumptions, not origin certificates. U.S. corporate domicile and a U.S. listing do not prove manufacturing origin. If the final content definition excludes the modeled products, their eligible fraction goes to zero.

For component and integrated suppliers, the filtered revenue pool means 3.2T-linked sales plus incremental cross-generation pricing spillover from competition for common qualified capacity. It is not a claim that 800G/1.6T is directly regulated. Fractions are LOW-confidence exposure proxies, not verified geographic market shares.

Signal ledger field	Current treatment
Signal ID / date	FCC-OPTICS-20261001; Serenity/MS source chain
What is already included	All saved product growth, shortages, capacity, customer probabilities and contracts. The bridge adds only the explicitly calculated incremental economics.
Probability and timing	40% common policy scenario; price/cost intensity 0 before Q3 2027, 25% Q3 2027, 50% Q4 2027, 75% Q1 2028, 100% from Q2 2028. AAOI mix uses its own slot ramp.
Numerical contribution	Q4 2027 operating quarter: delta revenue 2.210000, delta consolidated OP 1.989000 (USD millions). Forward-four-quarter valuation OP changes 16.339500.
Dependence	The policy probability is shared across companies. These are correlated portfolio exposures, not independent confirmations. Existing customer-conversion probabilities are already inside source revenue.
Raise weight	Rule/proposal scope supports qualifying content and timing; company product/foundry evidence improves; customer sourcing or like-for-like price negotiations support the incremental premium.
Reduce weight	Policy is delayed beyond the horizon, the content definition excludes products, supply eases, fixed contracts prevent repricing, unit costs absorb the premium, or 3.2T conversion slips.
Zero-contribution boundary	No broad TAM uplift, no new factory capacity, no OCS benefit and no blanket-ban recovery. AAOI receives no modeled units before 2028. MXL is assessed separately as a candidate.

Conditional inputs selected before calculating targets

Input	Value	Evidence status
CW relevant qualifying exposure	40.00%	LOW-confidence scenario allocation
UHP relevant qualifying exposure	60.00%	LOW-confidence scenario allocation
ELS relevant qualifying exposure	30.00%	LOW-confidence scenario allocation
Conditional incremental price premium	5.00%	LOW-confidence analyst assumption
Incremental COGS / price revenue	5.00%	LOW-confidence analyst assumption
Incremental opex / incremental revenue	5.00%	LOW-confidence analyst assumption

Incremental COGS and opex rates apply to incremental price revenue, not original product revenue or COGS. AAOI mix cost uses unit COGS separately. AXTI stranded COGS is a fraction of COGS attached to lost units. Chinese component-cost rates apply to affected reference COGS.

Timing is an analyst sensitivity. A deliberately labeled early-sourcing proxy ahead of a 2028 3.2T ramp. Earlier quarters receive zero incremental policy revenue. Full intensity begins Q2 2028 and remains an uncertain continuation through the saved forecast only. These dates are not announced orders or a published regulatory timetable. The probability sensitivity below includes a no-policy control; a later start would reduce the near-term contribution.

Incremental revenue, cost and profit

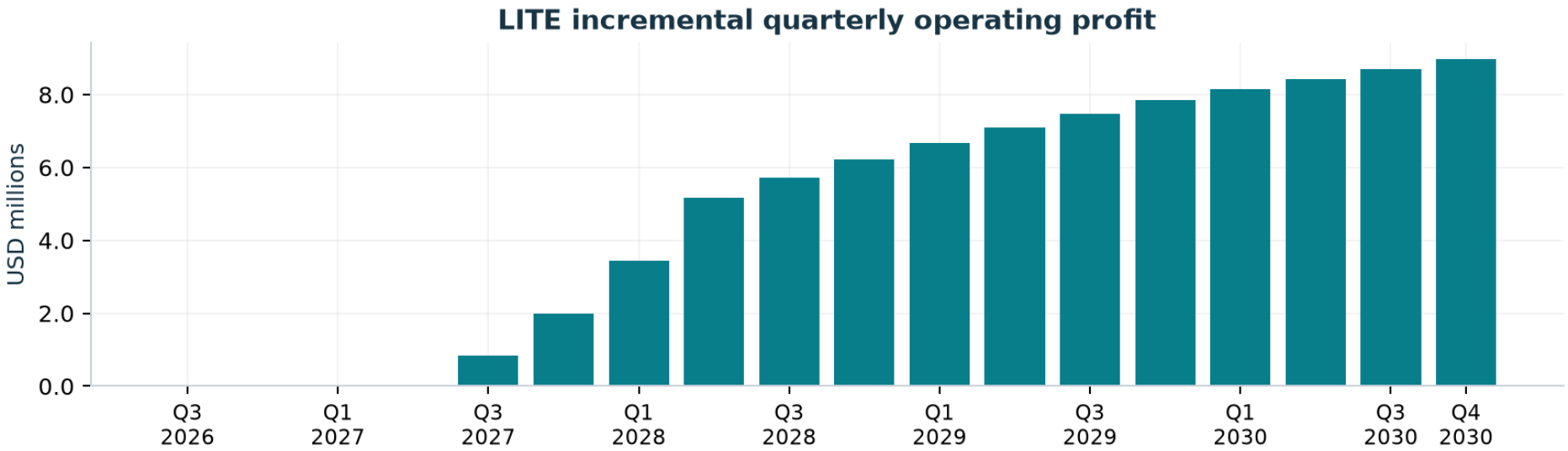


Figure 3. Incremental consolidated operating profit, USD millions. Positive bars show the weighted policy operating-profit increment; reference operations remain separate.

Quarter	Delta revenue	Delta COGS	Delta gross profit	Delta opex	Delta OP
Q3 2026	0.0000	0.0000	0.0000	0.0000	0.0000
Q4 2026	0.0000	0.0000	0.0000	0.0000	0.0000
Q1 2027	0.0000	0.0000	0.0000	0.0000	0.0000
Q2 2027	0.0000	0.0000	0.0000	0.0000	0.0000
Q3 2027	0.9425	0.0471	0.8954	0.0471	0.8483
Q4 2027	2.2100	0.1105	2.0995	0.1105	1.9890
Q1 2028	3.8250	0.1913	3.6338	0.1913	3.4425
Q2 2028	5.7500	0.2875	5.4625	0.2875	5.1750

Quarter	Delta revenue	Delta COGS	Delta gross profit	Delta opex	Delta OP
Q3 2028	6.3700	0.3185	6.0515	0.3185	5.7330
Q4 2028	6.9200	0.3460	6.5740	0.3460	6.2280
Q1 2029	7.4100	0.3705	7.0395	0.3705	6.6690
Q2 2029	7.9000	0.3950	7.5050	0.3950	7.1100
Q3 2029	8.3140	0.4157	7.8983	0.4157	7.4826
Q4 2029	8.7200	0.4360	8.2840	0.4360	7.8480
Q1 2030	9.0640	0.4532	8.6108	0.4532	8.1576
Q2 2030	9.3640	0.4682	8.8958	0.4682	8.4276
Q3 2030	9.6640	0.4832	9.1808	0.4832	8.6976
Q4 2030	9.9640	0.4982	9.4658	0.4982	8.9676

Active-window product bridge

Product / event	Pre-policy operating pool	Filtered exposure	Weighted delta revenue	Weighted delta OP
CW lasers	215.0000	86.0000	0.8600	0.7740
Ultra-high-power laser chips	215.0000	129.0000	1.2900	1.1610
ELS modules	20.0000	6.0000	0.0600	0.0540

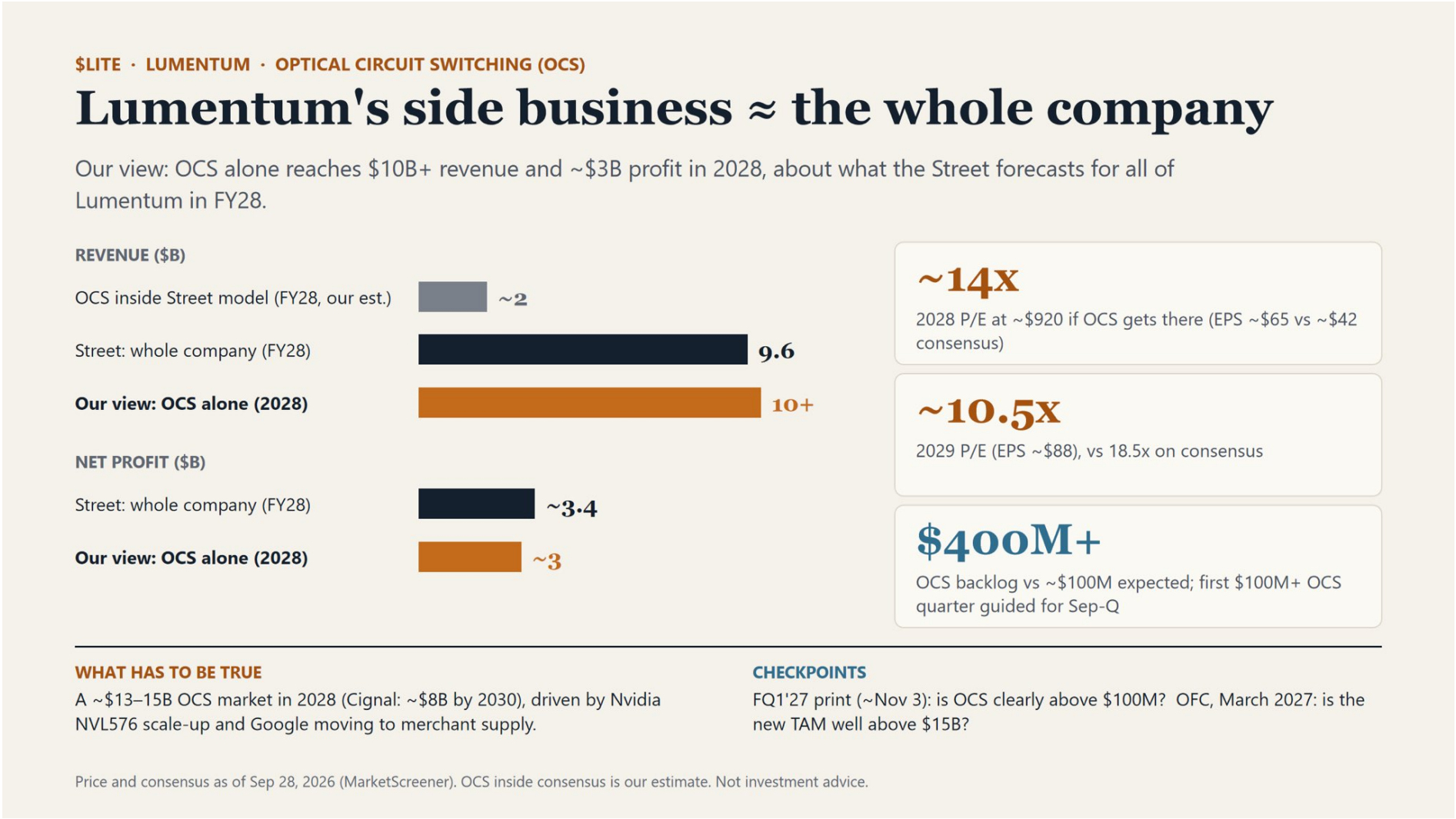
Filtered exposure is a scenario allocation, not verified U.S.-origin revenue. For cost cases the exposure is an affected cost pool; AAOI substitution units are explained in its separate bridge. All financial amounts use the company table units.

Business research and source evidence

The following sets out the company thesis, historical results, guidance, capacity, program probabilities, risks and original source images. Source dates remain visible. Operating assumptions below feed the complete current financial tables, including the explicit probability-weighted policy contribution.

Active OCS probability ledger

Hypothesis OCS-TWST-20260929: Lumentum reaches $10B of calendar-2028 OCS revenue, using the lower bound of TWST1000's $10B+ thesis. The post compares this with fiscal-year consensus, so calendar 2028 is a stated model interpretation. The $13-15B market claim implies approximately 67-77% Lumentum share. This is a LOW-confidence conditional outcome from a named source, not company guidance or a booked order.



User-supplied TWST1000 OCS thesis, September 29, 2026. $10B revenue, $3B net profit and consensus figures are source claims; CY/FY scopes differ.

Assigned probability: 30%. This is the midpoint of Rule 16b's 25-35% starting range for one credible source. The source supplies a concrete company revenue hypothesis and claims supply-chain access. Lumentum already has commercial OCS traction. The exact $10B scale and 2028 timing remain a single-source judgment; replies and repeated backlog references do not count as independent corroboration. The weight is subjective, selected from the evidence before computing the price target, not calibrated to a desired stock price.

The March OFC deck supports a $400M+ backlog, a multi-year multi-billion agreement and a >$1B 2027 run-rate objective. These are already represented in the reference OCS path and are not additional September orders. They support feasibility of growth, not certainty of the much larger endpoint. Greensboro's laser-chip

facility is not an OCS assembly-capacity figure.

Cignal's July forecast exceeds $8B in 2030 and includes GPU scale-up from 2028 and Google's merchant transition. Its Google vendor list includes Lumentum, Coherent and Huber+Suhner. The source case therefore requires both a larger, earlier market and strong Lumentum capture. Those differences lower the assigned probability; they do not make an unconfirmed forecast unusable. No source price target, $3B net-profit estimate or $65/$88 EPS claim replaces our operating-profit model.

Ledger field	Current treatment
Outcome probability	30% for the incremental $10B CY2028 OCS path; 70% retains reference economics in this simplified two-outcome bridge.
Reference already included	CY2028 OCS USD 1,527.0M, including existing commercial traction and modeled growth.
Conditional outcome	USD 10,000M in CY2028; named-source forecast endpoint. Ports, ASP, customer awards and capacity are unresolved cross-checks.
Increment and active contribution	10,000 - 1,527.0 = 8,473.0M conditional increment; 30% × 8,473.0 = 2,541.9M active increment. Active CY2028 OCS = 4,068.9M.
Workbook implementation	LITE_Q_Model B42 probability; B43 conditional endpoint; rows 52-69 incremental bridge; T19:T34 active OCS; existing OP and price target formulas retained.
Overlap/dependence	The backlog and agreement support the reference path. No separate order addition. Market size, capture and capacity are dependencies within this one hypothesis, not independent additive opportunities.
Probability escalation	Review toward 40-50% if independent sources or customer/management evidence specifically supports the larger scale and timing. Small initial orders alone do not validate $10B.
Probability reduction	Review toward 15-20% after material ramp misses, slower merchant outsourcing or weaker competitive capture. Remove the increment if the hypothesis is contradicted, not merely unconfirmed.

Quarterly timing and continuation assumptions

Q3-Q4 2026 remain at $180M/$220M OCS to preserve the existing earnings/guidance anchor. The conditional case interpolates geometrically from Q4 2026 to Q1 2028 at **59.79% quarterly growth** through the four 2027 quarters. This is an aggressive analyst-selected realization path, not a reported production schedule. The active model receives only 30% of its increment. CY2028 allocates $10B using the existing quarterly OCS mix; each quarter's conditional revenue is reference OCS × 6.548788.

For 2029-2030, the conditional path carries the higher realized level forward using the existing sequential OCS growth rates. This is a LOW-confidence continuation assumption, not another source forecast, a second TAM uplift or a new probability award. Removing the entire increment on January 1, 2029 would instead assume an unsupported collapse. Revisit the level, growth and probability as evidence arrives. The next 12-16 months receive the greatest analytical weight.

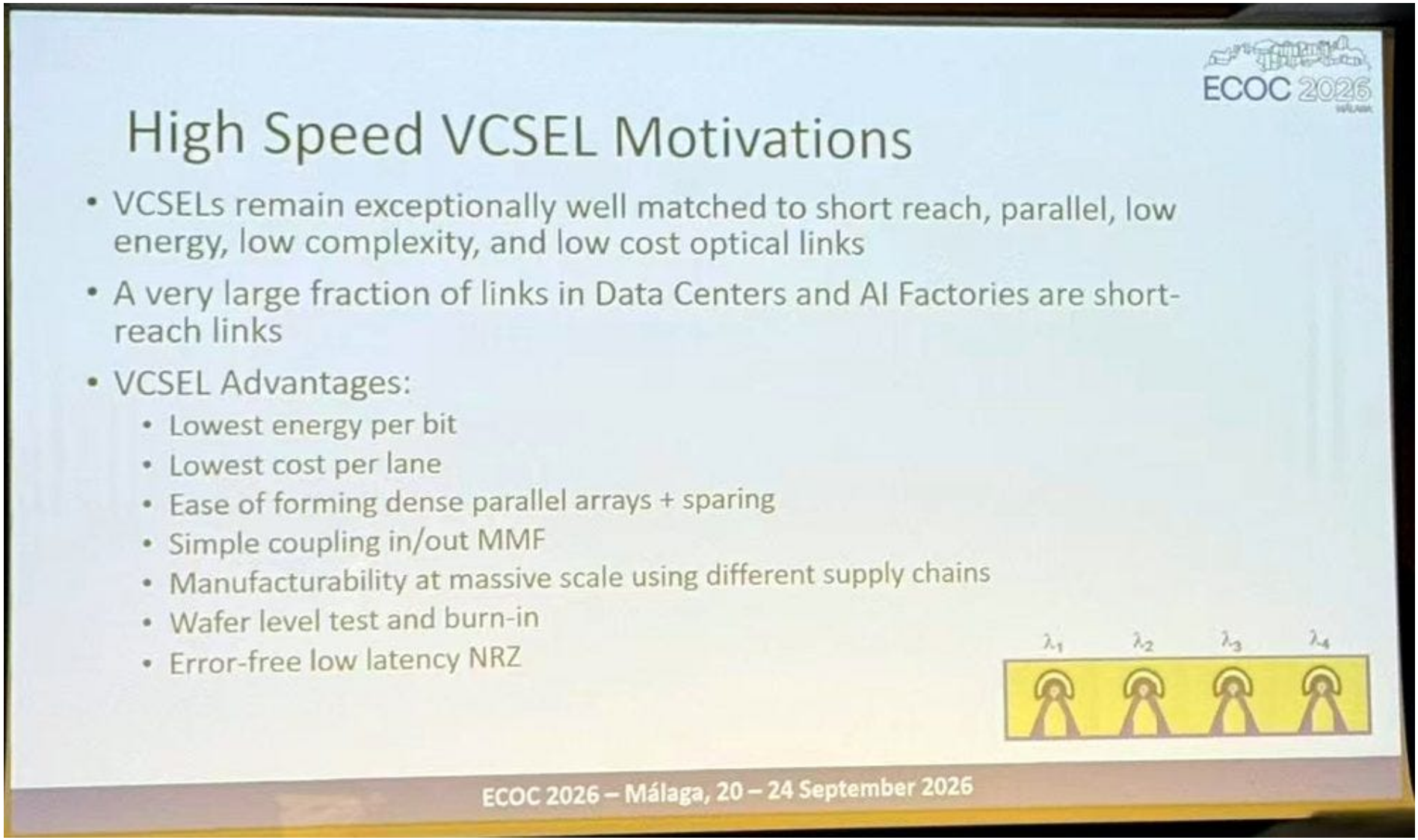
The reference and conditional OCS series feed one probability-weighted path. The complete operating model below shows the active quarterly financial tables.

OCS probability methodology

Only 30% is active. These controls recalculate the same OCS increment through unchanged product gross margins, opex ratios, 20x OP and shares. The scenario relationship is linear under these held-constant inputs. No Base increase is assigned merely for the same revenue opportunity: the existing TAM score is 9/10 and catalysts are 5/5.

Architecture evidence

Integrated and external light sources trade power, thermal coupling, repairability, packaging cost and laser utilization. Integrated NPO is not established as a temporary response to a shortage. GaAs/VCSEL scale-up is a possible Lumentum route, with different reach and density limits; it does not remove all InP demand. The ECOC demonstrations support feasibility rather than a particular volume forecast.



Public ECOC figure 1: VCSEL architecture discussion. Source: FUNDA free excerpt; technical claims are attributed, not a production forecast.

Public ECOC excerpt; Lumentum VCSEL demonstration.

Business and operating evidence

Earnings calibration: 2026-08-11 on FQ4 FY2026; supply evidence reviewed 2026-09-29 **Framework version:** 8.0; explicit OCS probability-weighted input, original company valuation method retained. **Status:** NOT HELD | ⚡ BOTTLENECK - EML laser chip supply **Artifact analytical Base:** 80% (48/60). **Portfolio Excel Base:** 99% (owner override). **Near-term share anchor:** 102.0M diluted shares for calendar Q3 2026; the active Q4 2027 price target uses 104.5M. DB1 CPO Photonics. **Multiple:** 20x forward-four-quarter operating profit (sanctioned company method) **Model tab:** LITE_Q_Model

Calendar mapping: Lumentum's fiscal year ends in late June. FQ4 FY2026 ended 2026-06-27 = calendar Q2 2026. FQ1 FY2027 = calendar Q3 2026.

1. Share count and structure

Near-term modeled diluted shares are 102.0M. Lumentum reported non-GAAP diluted weighted shares of 101.1M for FQ4 FY2026 and guides approximately 102M for FQ1 FY2027. The earlier working calibration used 77.80M basic shares and an approximately 23.3M basic-to-diluted gap for converts and equity compensation. That basic observation is superseded by the September 24 proxy counts in the October 7 reconciliation above; it is not a current outstanding-share figure.

The following historical conversion bridge belongs to the FQ4 calibration, not a new October issuance. Four series totalled $3,198.4M of principal at 2026-03-28: 2026 Notes $468.7M, 2028 Notes $861.0M, 2029 Notes $603.7M and 2032 Notes $1,265.0M. Lumentum equitised $1,125.0M during FQ4, which is 35.2% of the total and reconciles management's "approximately 35%" exactly. Roughly $2,073M remains outstanding, about 61% of it the 2032 series at a $187.77 conversion price at the calibration-date stock price. Lumentum holds $88.7M of capped calls on that series, which reduce economic dilution but are excluded from diluted EPS because they are anti-dilutive.

The price target divides operating profit by the diluted count. Operating profit is pre-interest and if-converted diluted shares assume the notes become equity and the interest disappears, so the two are consistent.

Reporting structure. Lumentum reports two product categories that sum to total revenue: Components $649.4M and Systems $356.9M in FQ4, so Components are 64.5% of revenue. Industrial lasers and cable access sit inside the Systems category.

2. Latest earnings anchor (AH-15)

FQ4 FY2026, quarter ended 2026-06-27, reported 2026-08-11 after the close.

Metric	FQ4 FY2026 actual
Revenue	$1,006.3M (+109% YoY)
- Components	$649.4M (+22% seq, +103% YoY)
- Systems	$356.9M (+30% seq, +123% YoY)
Non-GAAP gross margin	50.4% (+250bps seq, +1,260bps YoY)
Non-GAAP operating profit	**$368.8M, 36.6% margin** (+440bps seq, **+2,160bps YoY**)
Non-GAAP opex	$138.1M, 13.7% of revenue (SG&A $50.6M, R&D $87.5M)
Non-GAAP EPS	$3.23

Metric	FQ4 FY2026 actual
Adjusted EBITDA	$406.4M
Non-GAAP diluted shares	**101.1M**
GAAP gross margin	47.4%
GAAP operating margin	27.8%
Cash and short-term investments	$2.74B
Capex	$167M

Third consecutive quarter of greater than 20% sequential growth, eighth consecutive quarter of top-line growth.

Read the GAAP line carefully. GAAP net loss was $7.2B on a one-time non-cash charge of $7.8B from equitizing convertible notes. That is the accounting cost of removing $1.1B of debt while the stock is deeply in the money. It is non-cash and non-recurring, and it is also the same event that explains why the diluted count is 102M rather than 67M.

Non-GAAP gross margin crossed 50% against an internal target of reaching that threshold at a $2B quarterly run rate. It arrived at half the revenue level.

FQ1 FY2027 guidance

Revenue $1.225B to $1.275B, midpoint $1.25B, +130% YoY and another all-time record. Non-GAAP operating margin 39.5% to 40.5%. Non-GAAP diluted EPS $4.05 to $4.35. Tax rate 16.5%. Shares approximately 102M.

Management reached the $1.25B revenue target more than one quarter ahead of the plan outlined at OFC, and the guided operating margin exceeds the high end of the target model associated with that revenue level. Roughly half the sequential growth comes from Components and half from Systems.

At $2B quarterly revenue the OFC target model was 38% to 42% operating margin. The CFO on this call reframed it: treat 42% as the midpoint, with the range moving up 100 to 200 basis points. New formal targets at the next OFC.

4. Ten-line decomposed model (AH-11)

The reference workbook separates UHP chips and ELS modules. OCS includes the explicit 30% probability-weighted revenue increment described above. Lumentum reports Components and Systems; individual product allocations and margins remain LOW-confidence estimates. The current financial chapter carries the current product forecasts.

Gross profit is the sum of each product’s revenue times its own margin. Opex uses an explicit percentage schedule. Non-GAAP calibration is separate from forward assumptions.

5. Rolling forward-4Q price target (AH-1, AH-16)

price target(Q) = [OP(Q) + OP(Q+1) + OP(Q+2) + OP(Q+3)] x 20 / diluted shares

The ultra-high-power split (2026-08-11)

Irrational Analysis estimates **80-90% gross margin for ultra-high-power laser chips and 50-60% for ELS modules**. These are illustrative BOM and ASP assumptions, not disclosed Lumentum product margins. The model keeps chips and complete modules separate because their manufacturing content and economics differ. The example uses assumed wafer/die dimensions, yields, packaging costs and a guessed selling price; it is not an audited supplier cost sheet.

Product	100 mW CW	250 mW CW	650 mW SOA	400 mW CW (0.5-1 MHz linewidth) *competent vendor	400 mW CW (0.5-1 MHz linewidth) *loser vendor	200G EML	Coherent lite Rx
width (mm)	0.25	0.35	0.7	0.5	0.5	0.25	2
length (mm)	1	1.75	2	3	5	1.5	5
area (mm^2)	0.25	0.6125	1.4	1.5	2.5	0.375	10
wafer size (in)	4	4	4	4	4	4	6
dice per wafer	25888	10103	4551	4308	2526	17200	1513
Yield %	50	50	35	50	30	30	20
good dice	12944	5051.5	1592.85	2154	757.8	5160	302.6
Package yield %	50	50	40	50	50	50	80
sellable dice	6472	2525.75	637.14	1077	378.9	2580	242.08
pkg cost $	500	500	500	500	500	500	1000
wafer cost $	2000	2000	2000	2000	2000	2000	6000
process cost $	4000	4000	4000	4000	4000	4000	5000
COGS per dice $	1	2.6	10.2	6	17.2	2.5	49.6
ASP (guess) $	3.5	10	20	40	40	10	100
Gross Margin %	71	74	49	85	57	75	50
Optical Pow Per InP Area (mW/mm^2)	400	408.2	464.3	266.7	160		



Irrational Analysis, August 11: illustrative InP die and packaging cost model. Yield, die size, ASP and margin entries are analyst assumptions, not reported product results.

Management corroborates the direction of the comparison: the complete module sells for more than its constituent lasers, with margin above the corporate average but below the laser business. The excerpt supplies no exact 50-60% or 80-90% range.

Yes, Chris. Yes. No, we -- again, first, appreciate how much time you're spending on the company. Yes, we've got our first ELS module order right, which from an ASP standpoint is meaningfully higher than the set of lasers that we ship in. So we're very excited about the revenue opportunity, small to start, right? And we said that we'd be shipping sometime midyear, early second half of '27 to be honest, a little bit later.

We felt like we see more opportunity on sort of scale-out applications than we have to date. But we're very pleased to be participating in the early phases of scale up with both lasers and now with the ELS module. As I said a minute ago, the ELS module meaningfully higher from an ASP standpoint, but the margins are not quite as good, above corporate average but not quite as good as the lasers. So we're trading a little bit of margin to get some revenue bump. We think -- I think Simon asked earlier in the call, we think we have enough drivers on the margin side that we can keep the margin vector moving but help our revenue acceleration. .

Lumentum August earnings-call excerpt reproduced by Irrational Analysis. First ELS order and relative margin/ASP economics; exact product gross margins are not disclosed.

Margins are set at the midpoint of the independent BOM ranges: 85% for chips and 55% for modules as the anchor level. Before the explicit policy and scarcity price bridges, the reference shortage peak reaches 90.0% on chips, the top of the stated 80-90% range, and 59.0% on modules, just below the top of 50-60%. The complete current product tables include those bounded incremental price contributions. The source attributes much of the range to uncertain ASP. The 90% peak remains an aggressive model assumption within that estimated range, supported by scarcity but exposed to pricing, yield and customer-contract risk. It is not a measured margin or proof that competitors cannot qualify.

Lumentum received its **first** ELS module order this quarter, for 2H calendar 2027 delivery, so the line is all chip until then and becomes a mix afterward. The module mix is a visible drag on the family blend from 2028.

With ultra-high-power chips at an 84.0% anchor and the transceiver line at 26.9% in the calibration row, the model reproduces the disclosed 50.4% blend exactly. That is defensible on bought-in DSP content, and it is the disclosed blend working as the discipline it is meant to be.

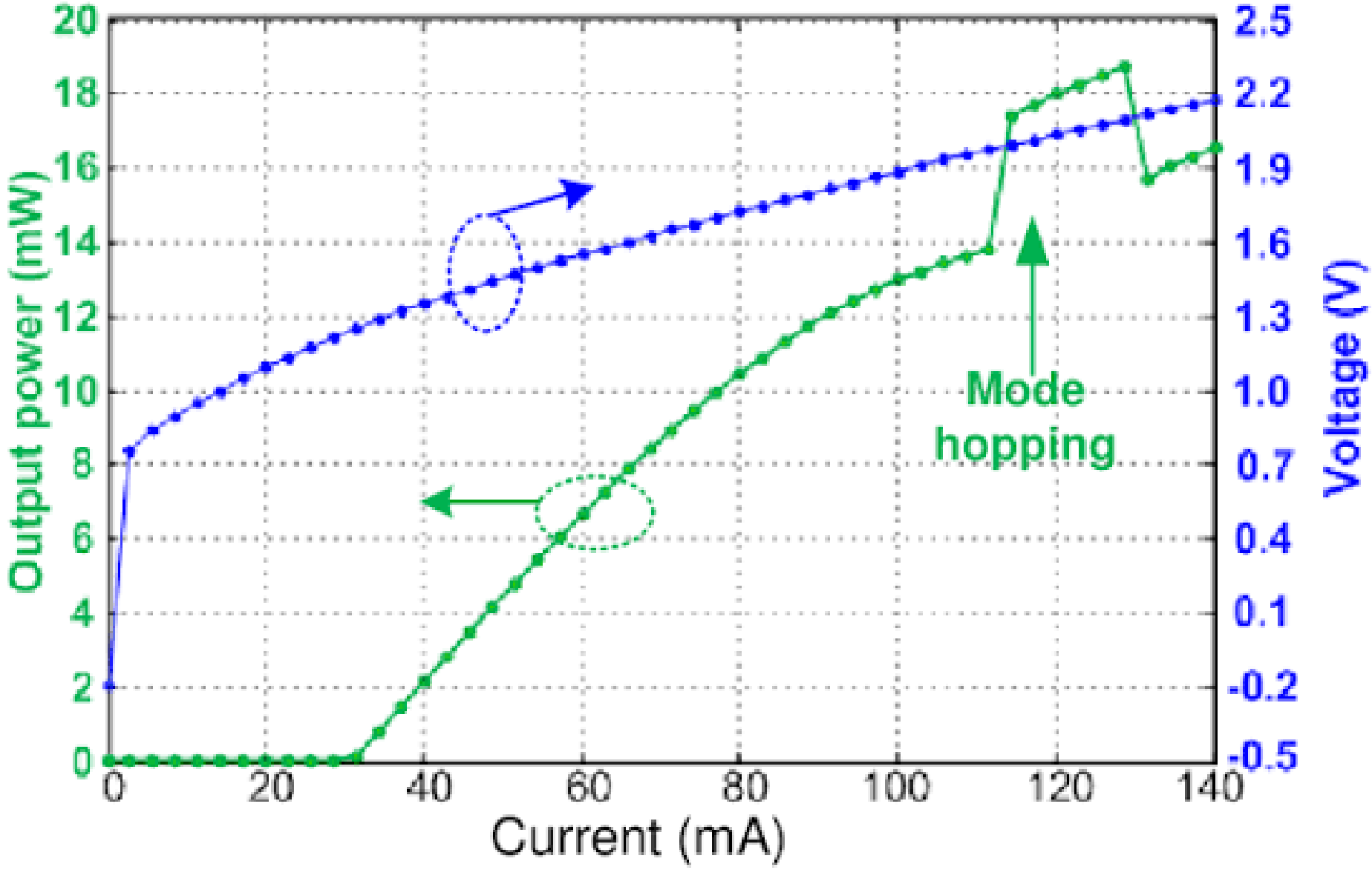
Margin shape: shortage peak 2027-2028, then give it back

Every laser line rises into the 2027-2028 shortage, peaks around Q1 to Q3 2028, then fades back to approximately its current level by Q4 2030.

The Systems lines end above where they start because their improvement is 1.6T ASP mix, OCS yield and scale rather than shortage pricing, and none of that reverses. The blend also ends above its reported anchor. UHP chips are 9.4% of modeled Q4 2030 revenue; the larger OCS mix now affects the derived blend. **Mix is permanent, the shortage premium is not.**

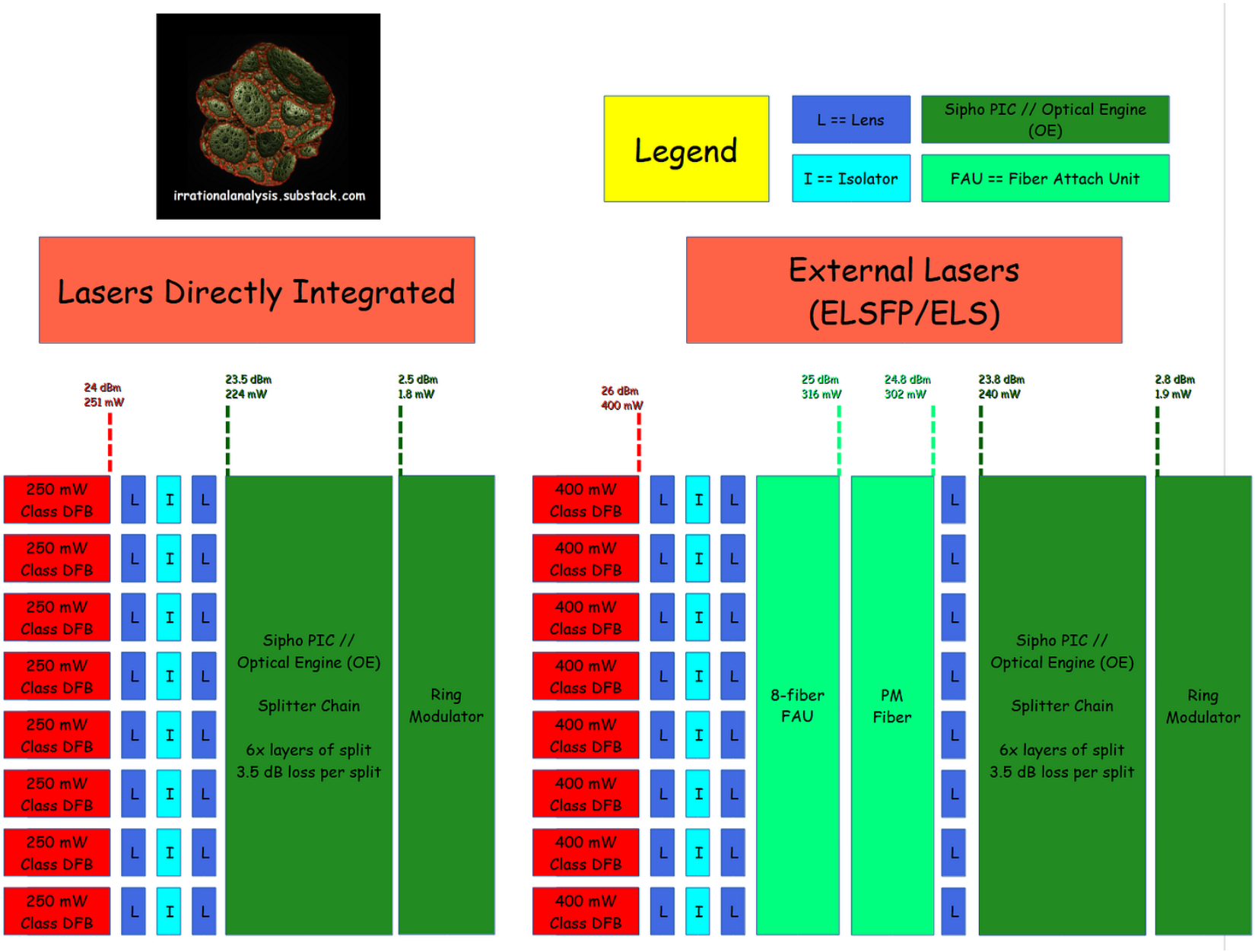
Qualification and mode-hop stability

A wider stable operating region across current and temperature can reduce customer qualification and link-reliability risk. Mode hops can disturb wavelength and power, with system impact depending on the optical link. Management discusses narrow specification spread and customer transceiver yields; the author's specific mode-hop explanation and roughly thousand-sample qualification threshold are interpretations, not issuer disclosures or a universal test standard.



Generic mode-hop illustration from the August 11 article. This low-power schematic explains a mechanism; it is not a measurement of Lumentum or Coherent UHP products.

Integrated versus external lasers: design-dependent tradeoffs



Irrational Analysis architecture example: integrated 250 mW sources versus external 400 mW sources. Losses and six splitter stages are specific assumptions; six binary stages mean 64 output branches.

Integration reduces some packaging and coupling burdens, while external sources can improve serviceability and isolate thermal/electrical noise. Power, reliability and utilization depend on the specific architecture. The ECOC evidence does not establish that integration will unwind when InP supply improves; retain both routes in the competitive assessment without an unsupported numerical reversal.

Wafer area and deliverable capacity

The author estimates roughly 65% relative area efficiency for a 400 mW versus 250 mW example, equivalent to about 1.54 times the wafer area per unit of optical power in that setup. This is an illustrative comparison, not a universal measured relationship. Saleable output depends on die size, edge losses, defect and packaging yield, qualification, cycle time and product allocation. The model does not infer a new unit-capacity limit from that ratio alone.

Wolfe Research, LLC

Congrats on all the success here. I guess I wanted to ask about indium phosphide substrate supply. I was really intrigued by the deal you guys signed in the quarter with AXTI. Because I think a quarter ago, you actually said that you were doing pretty well in indium phosphide substrate. And so it feels like there's a bit of a change here. I guess I'm wondering like what you guys are seeing longer term here in terms of your need for more indium phosphide substrate.

And as I extrapolate that kind of look at Greensboro, are you making more progress towards filling the rest of that facility looking forward?

Michael E. Hurlston
President, CEO & Director

George, yes, look, I would agree. I mean I think what we tried to hint to in a previous question is the ultrahigh-powered laser demand has surprised us. and that cuts across a couple of customers. So what we're trying to do, I think really in the last 3 months is secure even more substrate supply. We were doing well. We felt like we had given the baseline of demand that we were seeing from ultrahigh-powered and then, of course, the EMLs, the CW lasers that go into our scale-out products. We felt pretty good, right? -- as you said. But I think we've seen a pretty big surge in demand, as we said. And in order to respond to that, we went out and we found additional substrate help from AXTI the great partner, WooPenns worked with them for a good number of years. And we think we're going to need their help just given the surge in demand.

So that's what drove that deal. If this vector continues, we're probably going to need to look for more help on substrate we feel we're good at this moment. We're probably pretty comfortable just given our lead in arrangement with the Japanese supplier and now of the announced deal with AXTI. But given the rate of change in the demand vector that we're seeing, that may not be true a quarter or 2 from now either.

George Charles Notter
Wolfe Research, LLC

Got it. And then do you think it's likely that we could see you sign some LTAs for more of the capacity in Greensboro. Is that a possibility?

Michael E. Hurlston
President, CEO & Director

Yes. Look, we're looking to do that. I think we've sort of reported consistently that we have capacity in Greensboro. We have room there and Wupen has been out talking to a number of customers about helping them with their laser demands, which are, again, changing minute by minute is that everybody starts to look to adopt your packaged or co-packaged solutions. And so I would expect that we have some things to talk about over the next couple of quarters relative to new arrangements to in Greensboro that speak to more of this capacity we have there.

Lumentum August earnings-call excerpt on UHP demand, substrate supply and capacity. This is dated capacity context already represented in the model, not an additional October order.

Management's smaller-die discussion concerns CW lasers for 200G-per-lane silicon-photonics transceivers. It does not establish a smaller 400 mW UHP cavity with unchanged linewidth. No new UHP cost reduction is imported from that statement.

Michael E. Hurlston
President, CEO & Director

Okay, Chris, yes, good question, and I'll give some commentary, and then have Wupen comment as well. So one, no slowdown in our EML demand. I mean we are I think it was Papa that asked a minute ago, we see still a significant supply/demand imbalance on CW or EMLs. As Kathy put in the prepared remarks, we're still very much on track to increase our EML output year-over-year.

But even at the end of the year, we'd expect to be significantly behind demand. So no slowdown at all in the demand that we're seeing from EML. We acknowledge and we've said this before, that we would expect and are seeing CW lasers take a significant portion of transceiver output. Silicon photonics is a viable solution at 1.6t. Of course, we've also said, and it also seems to be bearing out that as we think about 3.2T, silicon photonics loses some of its advantages, and we'd expect to see EMLs come back in a meaningful way.

What I comment is a little bit of a shift in stance is that we are allocating some of our output now because we're actually doing quite a bit better in Japan than we expected in terms of output. And so what we're doing is part of that is allocating some of this excess output to CW lasers and we're now shipping CW lasers in a relatively meaningful way in this 200-gig per lane silicon photonics opportunity. What's also changed for us is we have significantly reduced the die size of our CW laser.

So before when you and I talked, we said, hey, probably from a margin standpoint, EML's better. Now we brought things much more in line. CW lasers are smaller. They're better performing. We are commanding a significant price premium, as I said in a previous question, against what we view as the market price because of the performance of these lasers. And as such, the margin opportunity is still better EMLs, but that gap has closed considerably, I think since the last time we talked. Wupen, do you want to talk a little bit about the dynamic that you're seeing on CW the EML?

Management discusses smaller CW dies and the 200G-per-lane opportunity. The product boundary matters: this is not a quantified UHP die-shrink disclosure.

Above-guidance disclosure (AH-15.5)

Q3 2026 revenue of $1,253M and operating margin of 39.70% remain within the guided revenue and operating-margin ranges. Later quarters are model assumptions. The OCS probability changes the revenue mix and hence the derived group margin, while product GM and opex/revenue inputs stay fixed.

Current Q4 2027 operating margin is 47.611%; the peak across the modeled horizon is 48.481%. These exceed the CFO's $2B target-model discussion of a 42% midpoint plus 100-200 basis points. The existing higher-margin chip and scale assumptions remain deliberate estimates. OCS's product gross margin is lower than the group's highest-margin chip lines, so increasing OCS does not mechanically increase the blended margin even though it increases profit dollars.

6. Product-level evidence from this call

EMLs. Record quarter. 200G per lane now over 25% of EML revenue, expected to exceed 50% of volume by mid-2027. Over 50% unit growth expected by the December 2026 quarter year over year. Still shipping behind demand with no change in the imbalance.

CW lasers. Die size significantly reduced, which improves margin per wafer. Management reports commanding a significant price premium against market price because the narrow spec distribution produces materially better transceiver yields for customers. The margin gap to EML has closed considerably. Excess Japanese output is being allocated to CW.

Pump lasers. Up over 80% year over year, effectively sold out for the foreseeable future, roughly 70-80% share. A fourfold shipment increase expected over the next several quarters. Multiple three-year customer agreements, mostly take-or-pay, with pricing levers and capex offset.

Narrow-linewidth assemblies. Tenth consecutive sequential growth quarter, up over 130% year over year.

OCS. First triple-digit revenue quarter guided in FQ1 and meaningfully above $100M. Tracking to roughly $400M for the second half of calendar 2026. Merchant supplier competing with Coherent and Huber+Suhner for relevant Google orders; not the sole industry supplier. Management expects to cross over to number one at the customer with an internal source in early 2027. Higher and lower port count products plus in-tray offerings added to the roadmap, all additive and landing in 2028.

Cloud Light transceivers. Bulk of shipments at 800G, 1.6T shipping as planned with uptake intensifying from FQ1 through calendar 2027. Management claims first to market ahead of larger competitors on several 1.6T designs.

ELS and NPO. First external light source module purchase order received for delivery in the second half of calendar 2027, ahead of customer scale-up deployments in 2028. Lead CPO customer plans on track with an increased demand signal. NPO described as completely additive, with the rest of the customer base prioritising it as an intermediate step. Laser family spans 120mW, 150mW and 400mW.

First ELS module order. Meaningfully higher ASP than the constituent lasers, margin above corporate average but below the laser line. Management describes it as trading a little margin for revenue.

Greensboro capacity and NVIDIA allocation

Lumentum’s issuer blog identifies up to USD 5B of annual Greensboro laser-chip revenue capacity, UHP/CW production and NVIDIA as a customer. This is facility capacity, not backlog or a recognized annual sales forecast. The existing August call timing remains the model reference: initial revenue early 2028 followed by ramping capacity. [L1]

Serenity’s September 29 digest attributes to Barclays an approximately 50% NVIDIA allocation and possible talks for more. The underlying note, original post timestamp and direct post link are unavailable. The allocation remains a credible attributed channel check. Half of USD 5B is USD 2.5B at full capacity, not USD 5B of

NVIDIA orders. It may overlap existing purchase commitments and the modeled chip ramp.

Separate ledger disposition - Greensboro allocation: the existing 35% provisional weight concerns the attributed allocation statement, not the $10B OCS outcome. A bounding proxy gives 50% × $5B = $2.5B at full facility utilization. That is a total-customer capacity exposure, not an increment to current UHP/ELS sales; attribution to products, utilization timing and overlap with already modeled commitments are unresolved. No separate dollar increment is added because the incremental scope cannot be isolated from the current chip/ELS path, not because the statement lacks official confirmation. Revisit when product/timing or commitment scope separates that increment. It is not OCS assembly capacity.

7. Key risks (each scored)

Risk	Score	Kill switch
Coherent's 6-inch InP. 2.25x geometric wafer area versus 4-inch, before edge losses, die size, yield and product qualification. Coherent reports improved six-inch yields; if qualified output scales, the component pricing power carrying this model erodes.	-1	Component gross margin below 55% without a mix explanation.
CPO and NPO timing. Management guides ELS deliveries to 2H calendar 2027 and scale-up deployments to 2028. Barclays argued in July that broad CPO adoption may take longer. An integration delay is partly self-hedging since traffic stays in EML-intensive pluggables, but a capex-driven delay hurts both.	-1	Scale-up guidance moving beyond 2028.
Customer concentration and NVIDIA entanglement. NVIDIA holds $2B of convertible preferred plus a purchase commitment, and is simultaneously a large customer and the architect of the CPO timing this model depends on.	-1	A top-two customer qualifying a second source at scale.
2028 capacity landing into renegotiated LTAs. Greensboro and expanded Japanese capacity arrive around FY27 LTA expiry. Commitments made at peak pricing against volumes that arrive later.	-1	FY28 LTAs renewed at lower unit pricing.
Total	**-4**	

Watch item, no additional scored deduction: Chinese EML competition. Asked directly about new Chinese indium phosphide fabs, management said it has seen no impact and does not expect any, citing differentiation on EMLs and high and mid-power lasers, and noting that several Chinese suppliers quoting large numbers are not delivering into the market today. This management assessment is counter-evidence to announced device capacity. Track qualified customer shipments and competitive pricing; announced capacity alone does not establish delivery at comparable specifications.

Counter-evidence held against the remaining four. Gross margin crossed 50% at half the revenue level originally targeted for that threshold. The demand-supply imbalance on EML is unchanged and has worsened on high-power lasers. Pump laser agreements are take-or-pay with pricing levers. The first ELS module order is a purchase order rather than a forecast.

8. Re-rate triggers

1. **FQ1 FY2027 report.** Pre-registered test: revenue at or above the $1.275B high end with operating margin at or above 40.5% confirms the slope. Revenue below $1.225B or margin below 39.5% cuts the 2027 path.

2. **Component gross margin, if ever disclosed separately.** The single largest remaining estimated input. The revenue split is now disclosed; the margin split is not.
3. **OCS probability and timing.** Track the 30% active path in the probability ledger. Compare actual revenue and forward delivery evidence with the explicit 2027 ramp; independent support for $10B scale/timing can justify 40-50%, while material misses or reduced capture can justify 15-20%. Repeated references to the existing $400M backlog do not escalate the weight.
4. **Ultra-high-power laser revenue.** Management: roughly $50M by end of calendar 2026, first triple-digit quarter in FQ3 FY2027. That is a dated, falsifiable milestone.
5. **Greensboro.** First revenue early 2028, ramping through 2028, full pitch by end 2028 into 2029. This remains a management timing reference.
6. **FY28 LTA terms.** Renewal at flat or higher pricing extends the component margin peak. Lower pricing deepens the fade.
7. **New OFC financial targets.** Management will issue them at the next OFC and is running ahead on every metric in the current set.

9. Cross-book read-throughs

- **AXTI.** Management confirmed on this call that the ultra-high-power laser demand surge drove a new substrate agreement with AXT, and said that if the vector continues they will probably need to look for more help on substrate. That is Lumentum stating its internal position across five InP fabs is insufficient. The strongest external validation of the AXTI thesis available, and it is now a direct management statement rather than an inference.
- **COHR.** Direct competitor on both vectors. The 6-inch InP line is the specific threat to the component margin.
- **SIVE.ST, held.** Competes at the CW and high-power laser layer for CPO and NPO, not in EMLs. Lumentum's laser family spans 120mW, 150mW and 400mW and it holds a first ELS module order for 2H calendar 2027. That is a dated, funded competitive position in the exact band the SIVE power-band question concerns, and it sharpens rather than resolves that open item.
- **TSEM.** Tower's insertion-loss advantage reduces the number of CW lasers per package. Lumentum's CW volume growth and Tower's content-per-engine argument point in opposite directions on laser units per module.
- **AAOI, held.** EML consumer and transceiver competitor.
- **IQE.L.** Epiwafer supply into the same InP chain.
- **MTSI.** Its 75mW CW laser targets production in late calendar 2027 into a market where Lumentum already holds an ELS order for the same window.
- **NVDA.** $2B convertible preferred plus purchase agreement, and the architecture decision that sets CPO timing for the whole optical complex.

10. Position and sources

L1 - Issuer laser-capacity discussion, dated June 25, 2026; retrieved September 29. Greensboro capacity context corroborates the facility scale; it does not verify Barclays allocation percentages.

D1 - Muse X digest received September 29, 2026, attributed to Serenity. Mirror https://twiscan.com/en/x/aleabitoreddit; no original timestamp, direct link or relevant image supplied. Channel checks remain distinct from signed disclosures.

The August earnings evidence anchors the unchanged 2026 calibration. The 30% OCS probability and explicit ramp/continuation assumptions are active forward model inputs. The current financial tables are synchronized to LITE_Q_Model in the master workbook.

OCS-TWST-20260929 - Tom-supplied TWST1000 (@X_____0x) post and image dated September 29, 2026. The conditional revenue endpoint, $13-15B TAM and supply-chain commentary are attributed source claims. No direct original post URL was supplied; the original image is embedded at its point of use. Probability,

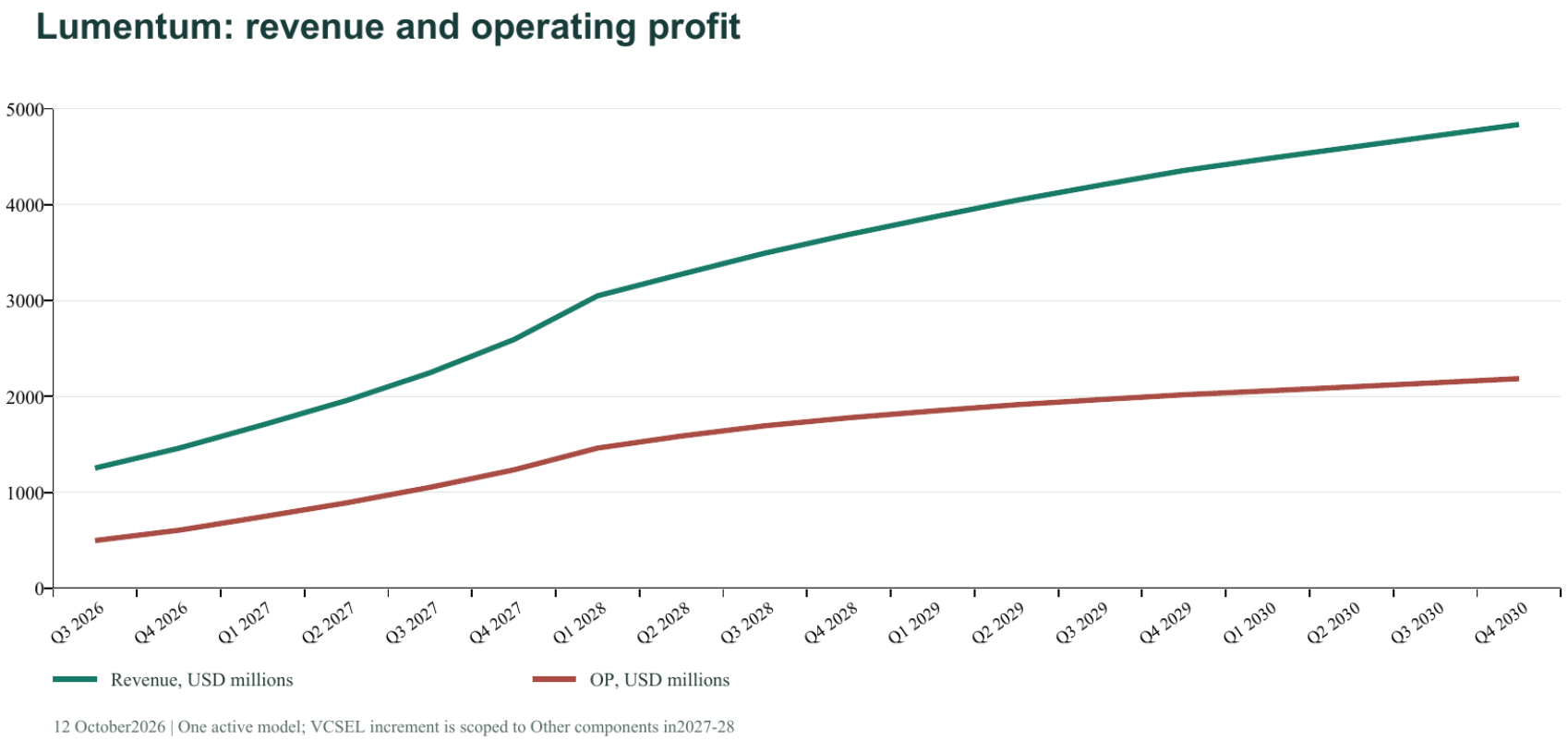
quarterly interpolation and subsequent continuation are our labeled modeling judgments. The 2026 OFC deck and Cignal report are linked in the probability section as support and counter-evidence, respectively.

OPTICA-20260924 - primary management appearance; captions reviewed October 2, 2026. Official Optica Day 1 recording and organizer agenda linked in the scarcity section. Question at 7:57:15; 2027 customer-request estimate at 7:57:46; balance expectation around 7:58:33. Captions, not separately transcribed audio, support this check. Product-specific quantities, request overlap and contract prices remain undisclosed.

LITE-PRICING-COMMENTARY-20261002 - Fabian and Zephyr. Original post links and overlap treatment appear beside the scarcity bridge. Opinions about licensing or stronger ASP are not disclosed contracts and add no separate revenue to the existing 55% premium case. Coherent qualification uncertainty remains uncertainty, not a failure finding.

Complete current operating model

All figures below include the OCS, FCC and qualified-laser scarcity contributions once, with the contract-timing reserve and the separate early merchant VCSEL branch in Other components. The VCSEL branch applies only to2027-28. USD millions unless labeled. Reported data and guidance retain their original dates; forward figures are analyst assumptions.



Lumentum: revenue and operating profit. USD millions. Model assumptions, not company guidance

Quarter	Revenue	COGS	Gross profit	Opex	OP	OP margin
Q3 2026	1,253.000	611.624	641.376	143.970	497.406	39.70%
Q4 2026	1,460.000	691.655	768.345	163.520	604.825	41.43%
Q1 2027	1,702.591	770.113	932.478	186.937	745.541	43.79%
Q2 2027	1,956.475	854.494	1,101.981	211.670	890.311	45.51%
Q3 2027	2,247.173	954.293	1,292.880	240.359	1,052.521	46.84%
Q4 2027	2,592.308	1,083.478	1,508.830	274.611	1,234.219	47.61%
Q1 2028	3,048.767	1,266.819	1,781.948	320.422	1,461.526	47.94%

Quarter	Revenue	COGS	Gross profit	Opex	OP	OP margin
Q2 2028	3,273.984	1,344.973	1,929.010	342.042	1,586.968	48.47%
Q3 2028	3,494.230	1,437.250	2,056.981	362.949	1,694.031	48.48%
Q4 2028	3,688.043	1,528.930	2,159.113	381.601	1,777.513	48.20%
Q1 2029	3,868.297	1,622.403	2,245.895	398.264	1,847.631	47.76%
Q2 2029	4,044.327	1,716.444	2,327.883	414.760	1,913.122	47.30%
Q3 2029	4,202.581	1,805.954	2,396.627	429.723	1,966.904	46.80%
Q4 2029	4,355.163	1,893.632	2,461.531	444.013	2,017.519	46.32%
Q1 2030	4,479.360	1,965.692	2,513.667	455.875	2,057.793	45.94%
Q2 2030	4,597.711	2,031.188	2,566.523	466.998	2,099.525	45.66%
Q3 2030	4,716.062	2,095.694	2,620.368	478.074	2,142.294	45.43%
Q4 2030	4,834.414	2,159.276	2,675.138	489.585	2,185.553	45.21%

Every revenue vector and product margin

Product gross profit equals product revenue × modeled product margin. Component allocations and margins are analyst assumptions unless separately identified as reported. These are the active financial paths; the preceding evidence sections distinguish dated source facts from model assumptions.

EML lasers

Quarter	Revenue	COGS	Gross margin	Gross profit
Q3 2026	300.000	100.500	66.50%	199.500
Q4 2026	335.000	108.875	67.50%	226.125
Q1 2027	365.000	113.880	68.80%	251.120
Q2 2027	395.000	118.500	70.00%	276.500
Q3 2027	425.000	123.250	71.00%	301.750
Q4 2027	455.000	128.765	71.70%	326.235
Q1 2028	490.000	137.200	72.00%	352.800
Q2 2028	520.000	145.600	72.00%	374.400
Q3 2028	550.000	156.750	71.50%	393.250
Q4 2028	575.000	167.900	70.80%	407.100
Q1 2029	600.000	180.000	70.00%	420.000
Q2 2029	625.000	192.500	69.20%	432.500
Q3 2029	645.000	203.820	68.40%	441.180
Q4 2029	665.000	215.460	67.60%	449.540
Q1 2030	680.000	224.400	67.00%	455.600
Q2 2030	695.000	232.130	66.60%	462.870
Q3 2030	710.000	239.270	66.30%	470.730
Q4 2030	725.000	246.500	66.00%	478.500

CW lasers

Quarter	Revenue	COGS	Gross margin	Gross profit
Q3 2026	90.000	34.650	61.50%	55.350
Q4 2026	115.000	43.125	62.50%	71.875
Q1 2027	140.000	50.680	63.80%	89.320
Q2 2027	165.000	57.750	65.00%	107.250
Q3 2027	190.380	64.619	66.06%	125.761
Q4 2027	215.860	71.423	66.91%	144.437
Q1 2028	241.440	78.552	67.47%	162.888
Q2 2028	267.120	86.231	67.72%	180.889
Q3 2028	292.320	95.236	67.42%	197.084
Q4 2028	312.480	103.664	66.83%	208.816
Q1 2029	332.640	112.992	66.03%	219.648
Q2 2029	352.800	122.640	65.24%	230.160
Q3 2029	370.944	131.891	64.44%	239.053
Q4 2029	388.080	140.679	63.75%	247.401
Q1 2030	401.184	147.817	63.15%	253.367
Q2 2030	413.280	154.324	62.66%	258.956
Q3 2030	425.376	160.529	62.26%	264.847
Q4 2030	437.472	166.830	61.87%	270.642

Ultra-high-power laser chips

Quarter	Revenue	COGS	Gross margin	Gross profit
Q3 2026	35.000	5.250	85.00%	29.750
Q4 2026	55.000	7.700	86.00%	47.300
Q1 2027	105.126	13.453	87.20%	91.674
Q2 2027	145.349	17.000	88.30%	128.349
Q3 2027	186.223	20.260	89.12%	165.963
Q4 2027	217.325	22.528	89.63%	194.797
Q1 2028	248.889	25.024	89.95%	223.866
Q2 2028	280.191	27.854	90.06%	252.337
Q3 2028	305.663	30.986	89.86%	274.676
Q4 2028	331.134	35.193	89.37%	295.941
Q1 2029	351.512	39.774	88.68%	311.738
Q2 2029	371.889	45.000	87.90%	326.889
Q3 2029	387.173	49.889	87.11%	337.283
Q4 2029	402.456	54.624	86.43%	347.832
Q1 2030	416.329	58.607	85.92%	357.722
Q2 2030	428.515	61.588	85.63%	366.926
Q3 2030	440.700	64.208	85.43%	376.492
Q4 2030	452.885	66.429	85.33%	386.456

External light source modules

Quarter	Revenue	COGS	Gross margin	Gross profit
Q3 2026	0.000	0.000	55.00%	0.000
Q4 2026	0.000	0.000	55.00%	0.000
Q1 2027	0.000	0.000	55.00%	0.000
Q2 2027	0.000	0.000	55.00%	0.000
Q3 2027	5.026	2.232	55.58%	2.793
Q4 2027	20.156	8.773	56.48%	11.384
Q1 2028	40.455	17.157	57.59%	23.298
Q2 2028	55.708	23.154	58.44%	32.554
Q3 2028	75.966	31.124	59.03%	44.842
Q4 2028	91.159	36.989	59.42%	54.170
Q1 2029	106.352	43.364	59.23%	62.988
Q2 2029	121.545	50.038	58.83%	71.507
Q3 2029	136.738	56.968	58.34%	79.770
Q4 2029	151.931	64.198	57.75%	87.733
Q1 2030	161.510	69.223	57.14%	92.287
Q2 2030	171.604	74.399	56.64%	97.205
Q3 2030	181.699	79.496	56.25%	102.203
Q4 2030	191.793	84.672	55.85%	107.121

Pump lasers

Quarter	Revenue	COGS	Gross margin	Gross profit
Q3 2026	180.000	74.700	58.50%	105.300
Q4 2026	210.000	85.050	59.50%	124.950
Q1 2027	240.000	94.320	60.70%	145.680
Q2 2027	268.000	102.376	61.80%	165.624
Q3 2027	295.000	110.035	62.70%	184.965
Q4 2027	320.000	117.120	63.40%	202.880
Q1 2028	345.000	124.890	63.80%	220.110
Q2 2028	368.000	132.480	64.00%	235.520
Q3 2028	390.000	141.570	63.70%	248.430
Q4 2028	410.000	151.290	63.10%	258.710
Q1 2029	428.000	160.928	62.40%	267.072
Q2 2029	445.000	170.435	61.70%	274.565
Q3 2029	460.000	179.400	61.00%	280.600
Q4 2029	475.000	188.575	60.30%	286.425
Q1 2030	487.000	196.261	59.70%	290.739
Q2 2030	498.000	202.686	59.30%	295.314
Q3 2030	509.000	209.199	58.90%	299.801
Q4 2030	520.000	215.280	58.60%	304.720

Narrow-linewidth assemblies

Quarter	Revenue	COGS	Gross margin	Gross profit
Q3 2026	110.000	47.850	56.50%	62.150
Q4 2026	125.000	53.250	57.40%	71.750
Q1 2027	138.000	57.270	58.50%	80.730
Q2 2027	150.000	60.750	59.50%	89.250
Q3 2027	162.000	64.152	60.40%	97.848
Q4 2027	173.000	67.297	61.10%	105.703
Q1 2028	185.000	71.040	61.60%	113.960
Q2 2028	196.000	74.872	61.80%	121.128
Q3 2028	207.000	79.488	61.60%	127.512
Q4 2028	217.000	84.413	61.10%	132.587
Q1 2029	226.000	89.270	60.50%	136.730
Q2 2029	235.000	94.235	59.90%	140.765
Q3 2029	243.000	98.901	59.30%	144.099
Q4 2029	251.000	103.663	58.70%	147.337
Q1 2030	258.000	107.844	58.20%	150.156
Q2 2030	265.000	111.830	57.80%	153.170
Q3 2030	272.000	115.872	57.40%	156.128
Q4 2030	279.000	119.970	57.00%	159.030

Other components

Quarter	Revenue	COGS	Gross margin	Gross profit
Q3 2026	56.000	34.440	38.50%	21.560
Q4 2026	60.000	36.600	39.00%	23.400
Q1 2027	62.000	37.386	39.70%	24.614
Q2 2027	65.000	38.805	40.30%	26.195
Q3 2027	68.150	40.349	40.79%	27.801
Q4 2027	70.450	41.438	41.18%	29.012
Q1 2028	73.360	42.926	41.49%	30.434
Q2 2028	76.720	44.749	41.67%	31.971
Q3 2028	80.080	46.799	41.56%	33.281
Q4 2028	83.440	48.939	41.35%	34.501
Q1 2029	85.000	50.065	41.10%	34.935
Q2 2029	88.000	52.096	40.80%	35.904
Q3 2029	91.000	54.145	40.50%	36.855
Q4 2029	94.000	56.212	40.20%	37.788
Q1 2030	97.000	58.200	40.00%	38.800
Q2 2030	100.000	60.200	39.80%	39.800
Q3 2030	103.000	62.212	39.60%	40.788
Q4 2030	106.000	64.236	39.40%	41.764

Cloud Light transceivers

Quarter	Revenue	COGS	Gross margin	Gross profit
Q3 2026	240.000	173.040	27.90%	66.960
Q4 2026	275.000	194.975	29.10%	80.025
Q1 2027	305.000	211.975	30.50%	93.025
Q2 2027	335.000	228.470	31.80%	106.530
Q3 2027	362.000	242.902	32.90%	119.098
Q4 2027	388.000	256.856	33.80%	131.144
Q1 2028	415.000	271.825	34.50%	143.175
Q2 2028	440.000	286.440	34.90%	153.560
Q3 2028	464.000	302.064	34.90%	161.936
Q4 2028	486.000	317.358	34.70%	168.642
Q1 2029	506.000	331.936	34.40%	174.064
Q2 2029	525.000	345.975	34.10%	179.025
Q3 2029	543.000	359.466	33.80%	183.534
Q4 2029	560.000	372.400	33.50%	187.600
Q1 2030	574.000	382.858	33.30%	191.142
Q2 2030	588.000	393.372	33.10%	194.628
Q3 2030	602.000	403.942	32.90%	198.058
Q4 2030	616.000	413.952	32.80%	202.048

Optical circuit switching

Quarter	Revenue	COGS	Gross margin	Gross profit
Q3 2026	180.000	101.700	43.50%	78.300
Q4 2026	220.000	121.000	45.00%	99.000
Q1 2027	280.464	149.207	46.80%	131.257
Q2 2027	363.126	187.373	48.40%	175.753
Q3 2027	481.395	242.142	49.70%	239.253
Q4 2027	658.517	323.991	50.80%	334.527
Q1 2028	932.623	451.389	51.60%	481.233
Q2 2028	991.245	475.798	52.00%	515.447
Q3 2028	1,047.202	503.704	51.90%	543.498
Q4 2028	1,097.830	532.448	51.50%	565.383
Q1 2029	1,145.794	561.439	51.00%	584.355
Q2 2029	1,191.093	589.591	50.50%	601.502
Q3 2029	1,233.727	615.630	50.10%	618.097
Q4 2029	1,273.696	640.669	49.70%	633.027
Q1 2030	1,308.337	662.018	49.40%	646.318
Q2 2030	1,340.312	680.879	49.20%	659.434
Q3 2030	1,372.288	699.867	49.00%	672.421
Q4 2030	1,404.263	718.983	48.80%	685.281

Industrial systems

Quarter	Revenue	COGS	Gross margin	Gross profit
Q3 2026	62.000	39.494	36.30%	22.506
Q4 2026	65.000	41.080	36.80%	23.920
Q1 2027	67.000	41.942	37.40%	25.058
Q2 2027	70.000	43.470	37.90%	26.530
Q3 2027	72.000	44.352	38.40%	27.648
Q4 2027	74.000	45.288	38.80%	28.712
Q1 2028	77.000	46.816	39.20%	30.184
Q2 2028	79.000	47.795	39.50%	31.205
Q3 2028	82.000	49.528	39.60%	32.472
Q4 2028	84.000	50.736	39.60%	33.264
Q1 2029	87.000	52.635	39.50%	34.365
Q2 2029	89.000	53.934	39.40%	35.066
Q3 2029	92.000	55.844	39.30%	36.156
Q4 2029	94.000	57.152	39.20%	36.848
Q1 2030	96.000	58.464	39.10%	37.536
Q2 2030	98.000	59.780	39.00%	38.220
Q3 2030	100.000	61.100	38.90%	38.900
Q4 2030	102.000	62.424	38.80%	39.576

Full price target derivation

The earnings window begins in the named quarter and includes its next three quarters. price target = summed operating profit × 20 / the saved quarter-specific diluted shares. No Base probability is applied again.

price target quarter / end	F4Q OP	Multiple	Shares M	USD/share
Q3 2026 / Q2 2027	2,738.083	20x	102.000	536.879024

price target quarter / end	F4Q OP	Multiple	Shares M	USD/share
Q4 2026 / Q3 2027	3,293.198	20x	102.500	642.575210
Q1 2027 / Q4 2027	3,922.592	20x	103.000	761.668346
Q2 2027 / Q1 2028	4,638.577	20x	103.500	896.343430
Q3 2027 / Q2 2028	5,335.234	20x	104.000	1,026.006625
Q4 2027 / Q3 2028	5,976.745	20x	104.500	1,143.874580
Q1 2028 / Q4 2028	6,520.038	20x	105.000	1,241.912067
Q2 2028 / Q1 2029	6,906.143	20x	105.500	1,309.221415
Q3 2028 / Q2 2029	7,232.297	20x	106.000	1,364.584426
Q4 2028 / Q3 2029	7,505.170	20x	106.500	1,409.421558
Q1 2029 / Q4 2029	7,745.176	20x	107.000	1,447.696392
Q2 2029 / Q1 2030	7,955.338	20x	107.500	1,480.062804
Q3 2029 / Q2 2030	8,141.740	20x	108.000	1,507.729671
Q4 2029 / Q3 2030	8,317.131	20x	108.500	1,533.111681
Q1 2030 / Q4 2030	8,485.165	20x	109.000	1,556.911052

Q4 2027: (1234.219027 + 1461.526339 + 1586.967853 + 1694.031463) ×20 /104.5 = **1,143.874580 USD/share**.

Annual cross-check

Calendar year	Coverage	Revenue	Gross profit	OP	OP margin
2026	2 quarters	2,713.000	1,409.721	1,102.231	40.63%
2027	4 quarters	8,498.547	4,836.169	3,922.592	46.16%
2028	4 quarters	13,505.024	7,927.052	6,520.038	48.28%
2029	4 quarters	16,470.369	9,431.936	7,745.176	47.02%
2030	4 quarters	18,627.547	10,375.697	8,485.165	45.55%

Probability and dilution sensitivity

Only one selected case is active. The following are diagnostic alternatives; OCS stays 30% and FCC stays 40% when varying scarcity probability.

Scarcity outcome probability	Q4 2027 USD/share
0.00%	1,142.5834
25.00%	1,143.1703
55.00%	1,143.8746
75.00%	1,144.3441
100.00%	1,144.9310

FCC probability; scarcity fixed at55%	Q4 2027 USD/share
0.00%	1,140.7474
20.00%	1,142.3110
40.00%	1,143.8746
60.00%	1,145.4382
100.00%	1,148.5653

Additional diluted shares	Q4 2027 USD/share
0.00%	1,143.8746
5.00%	1,089.4044
10.00%	1,039.8860

Validation and next evidence

Current model: LITE_Q_Model in v3_2_master_portfolio.xlsx. The selected55% scarcity,40% policy,30% OCS and60% early merchant VCSEL outcomes enter their defined operating pools exactly once. The VCSEL branch includes50% baseline overlap and6pp incremental cost buffers, with no extra branch beyond2028. Quarter convention, 20× multiple, shares, all unrelated vectors and Portfolio Excel Base 99 remain unchanged. The zero-scarcity row is a diagnostic control with the defined scarcity increment removed; it is not a second active case.

Revisit the exposed fraction, 5% premium and timing when accepted contract prices, product-specific unit and request denominators, allocation terms or unit-cost/yield evidence change the economic bridge. Lower shortage persistence, longer fixed-price commitments or a later 3.2T ramp reduce the increment. Official confirmation is

not a prerequisite to modeling; independent economic scope and avoidance of duplicate growth are the tests. Confidence declines beyond 12-16 months.

October 7 evidence sources

- **LITE-LTA1** · October 6, 2026, 18:18 UTC · P Equity Research laser-supply post. DOTX-20261006-1530 revision 2, post U305. Negotiation/pricing claim; logo-only image adds no commercial details.
- **LITE-PROXY1** · Filed October 6, 2026; record date September 24 · Definitive proxy. Common/preferred counts, conversion/voting rights and award/plan-reserve scope. Saved original read from DOT-EARNINGS-20261006T1924 revision 1.
- **LITE-ARS1** · October 6, 2026 annual-report republication · Annual report and fiscal 2026 Form 10-K. August 28 basic-common observation. Republication is not a new quarter or another recognition of previously disclosed conversion expense.